Solutions to the exercises in *Categorical Logic and Type Theory* by Bart Jacobs

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Contents

Contents **i**

0 Prospectus **1**
0.2. The logic and type theory of sets, 0.2.1, 0.2.2, 0.2.3, 0.2.4

1 Introduction to fibred category theory **6**
1.1. Fibrations, 1.1.1, 1.1.2, 1.1.3, 1.1.4, 1.1.5, 1.1.6, 1.1.7, 1.1.8, 1.1.9, 1.1.10, *1.2. Some concrete examples: sets, ω-sets and PERs*, 1.2.1, 1.2.2, 1.2.3, 1.2.4, 1.2.5, 1.2.6, 1.2.7, 1.2.8, *1.3. Some general examples*, 1.3.1, *1.6. Fibrations of signatures*, 1.6.1, 1.6.2, 1.6.3, 1.6.4, 1.6.5, 1.6.6, 1.6.7

Bibliography **47**

Chapter 0

Prospectus

0.2 The logic and type theory of sets

0.2.1. Define a left adjoint $F : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Pred}$ to the inclusion functor

$$\begin{tikzcd}
\mathbf{Pred} \arrow[rr, hook] \arrow[dr] & & \mathrm{Fam}(\mathbf{Sets}) \arrow[ll, bend right, "F"'] \arrow[dl] \\
& \mathbf{Sets} &
\end{tikzcd}$$

such that: (1) F makes the triangle commute (so it does not change the index set), and (2) F commutes with substitution.

Solution. Recall that the inclusion functor $G : \mathbf{Pred} \hookrightarrow \mathrm{Fam}(\mathbf{Sets})$ sends a predicate (I, X) (where $X \subseteq I$) to the family $G(I, X) = (X_i)_{i \in I}$, where

$$X_i = \begin{cases} \{*\} & \text{if } i \in X \\ \emptyset & \text{otherwise.} \end{cases}$$

Define the functor $F : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Pred}$ on objects by

$$F\left((X_i)_{i \in I}\right) = \left(I, \left\{i \in I : X_i \neq \emptyset\right\}\right)$$

and on morphisms by

$$F\left((X_i)_{i \in I} \xrightarrow{(u,f)} (Y_j)_{j \in J}\right) = u.$$

Note that F is well-defined on morphisms. Indeed, suppose $(X_i)_{i \in I}$ and $(Y_j)_{j \in J}$ are families of sets and $(u, f) : (X_i)_{i \in I} \to (Y_j)_{j \in J}$ is a morphism of families. If $i \in I$ and $X_i \neq \emptyset$, then $\emptyset \neq f(X_i) \subseteq Y_{u(i)}$, so $Y_{u(i)} \neq \emptyset$, and hence

u is a morphism from $F\left((X_i)_{i\in I}\right)$ to $F\left((Y_j)_{j\in J}\right)$ in **Pred**. Moreover, F does not change the index set by definition.

Next, we claim that $F \dashv G$ by demonstrating the adjointness criteria in [2, Definition 3.1.4], namely that there is a "unit" natural transformation

$$\eta : \mathrm{id}_{\mathrm{Fam}(\mathbf{Sets})} \longrightarrow G \circ F.$$

such that for each I-indexed family of sets $(X_i)_{i\in I}$, each object (J, Y) in **Pred**, and each morphism $(v, g) : (X_i)_{i\in I} \to G(J, Y)$ in Fam(**Sets**), there exists a unique morphism $v' : F\left((X_i)_{i\in I}\right) \to (J, Y)$ in **Pred** such that $G(v') \circ \eta_{(X_i)_{i\in I}} = (v, g)$, as in the following commutative diagram.

$$\begin{tikzcd}
\textbf{Pred} & \mathrm{Fam}(\textbf{Sets}) \\
F\left((X_i)_{i\in I}\right) \arrow[d, dashed, "v'"] & (X_i)_{i\in I} \arrow[r, "\eta_{(X_i)_{i\in I}}"] \arrow[dr, "(v,g)"'] & G\left(F\left((X_i)_{i\in I}\right)\right) \arrow[d, dashed, "G(v')"] \\
(J, Y) & & G(J, Y)
\end{tikzcd}$$

To this end, for an object $(X_i)_{i\in I}$ define the component

$$\eta_{(X_i)_{i\in I}} : (X_i)_{i\in I} \longrightarrow G\left(F\left((X_i)_{i\in I}\right)\right) = \left(\text{if } X_i \neq \emptyset \text{ then } \{*\} \text{ else } \emptyset\right)_{i\in I}$$

as

$$\eta_{(X_i)_{i\in I}} = \left(\mathrm{id}_I, \left(f_i : X_i \longrightarrow \text{if } X_i \neq \emptyset \text{ then } \{*\} \text{ else } \emptyset\right)_{i\in I}\right),$$

where

$$f_i = \begin{cases} x \mapsto * & \text{if } X_i \neq \emptyset \\ \mathrm{id}_\emptyset & \text{otherwise.} \end{cases}$$

We omit the rest of the verification that η is the unit of the adjunction $F \dashv G$.

Finally, we claim that F commutes with substitution. Indeed, if $u : I \to J$ is a morphism in **Sets** and $(Y_i)_{j\in J}$ is a J-indexed family of sets, then

$$\begin{aligned}
u^*\left(F\left((Y_j)_{j\in J}\right)\right) &= u^*\left(J, \left\{j \in J : Y_j \neq \emptyset\right\}\right) \\
&= \left(I, \left\{i \in I : Y_{u(i)} \neq \emptyset\right\}\right) \\
&= F\left(\left(Y_{u(i)}\right)_{i\in I}\right) \\
&= F\left(u^*\left((Y_j)_{j\in J}\right)\right).
\end{aligned}$$

□

0.2.2. Define for a subset $X \subseteq I$ the relation $\mathrm{nEq}(X) \subseteq I \times I$ by

$$\mathrm{nEq}(X) = \Big\{(i, i') : i \neq i' \text{ or } i \in X\Big\}$$

and show that the assignment $X \mapsto \mathrm{nEq}(X)$ is right adjoint to contraction $\delta^* : P(I \times I) \to P(I)$. Notice that $\mathrm{nEq}(X)$ at the bottom element $X = \emptyset$ is inequality on I.

Solution. Fix a set I and consider the functors

$$P(I) \underset{\mathrm{nEq}}{\overset{\delta^*}{\leftrightarrows}} P(I \times I).$$

To show that $\delta^* \dashv \mathrm{nEq}$, we fix $X \subseteq I$ and $Y \subseteq I \times I$ and claim that $\delta^*(Y) \subseteq X$ if and only if $Y \subseteq \mathrm{nEq}(X)$. Suppose $\delta^*(Y) \subseteq X$ and $(i, j) \in Y$. If $i \neq j$, then $(i, j) \in \mathrm{nEq}(X)$, and if $i = j$, then $i \in \delta^*(Y) \subseteq X$, and hence $(i, j) \in \mathrm{nEq}(X)$. Thus, $Y \subseteq \mathrm{nEq}(X)$. Conversely, suppose $Y \subseteq \mathrm{nEq}(X)$ and $i \in \delta^*(Y)$. Then $(i, i) \in Y \subseteq \mathrm{nEq}(X)$, so $i \in X$, whence $\delta^*(Y) \subseteq X$ and $(i, j) \in Y$. $\square$

0.2.3. Show that the equality functor $\mathrm{Eq} : \mathbf{Sets} \to \mathbf{Rel}$ also has a right adjoint.

Solution. Define the functor

$$F : \mathbf{Rel} \longrightarrow \mathbf{Sets}$$

on objects by

$$F(I, R) = \{i \in I : (i, i) \in R\}$$

and on morphisms by

$$F\Big((I, R) \xrightarrow{u} (J, S)\Big) = u|_{F(I,R)}.$$

Note that F is well-defined on morphisms since if $u : (I, R) \to (J, S)$ is a morphism in $\mathbf{Rel}$ and $i \in F(I, R)$, then $(i, i) \in R$, so $(u(i), u(i)) \in S$, and hence $u(i) \in F(J, S)$. We claim that $\mathrm{Eq} \dashv F$. Fix an object I in $\mathbf{Sets}$ and an object (J, S) in $\mathbf{Rel}$. Define functions

$$f : \mathbf{Rel}(\mathrm{Eq}(I), (J, S)) \longrightarrow \mathbf{Sets}(I, F(J, S))$$
$$g : \mathbf{Sets}(I, F(J, S)) \longrightarrow \mathbf{Rel}(\mathrm{Eq}(I), (J, S))$$

by

$$f\left(\mathrm{Eq}(I) \xrightarrow{u} (J,S)\right) = u \qquad\qquad g\left(I \xrightarrow{u} F(J,S)\right) = u.$$

Note that f is well-defined since if $u : \mathrm{Eq}(I) \to (J,S)$ is a morphism in **Rel** and $i \in I$, then $(i,i) \in \mathrm{Eq}(I)$, so $(u(i), u(i)) \in S$, and hence $u(i) \in F(J,S)$. Similarly, note that g is well-defined since if $u : I \to F(J,S)$ is a morphism in **Sets** and $i \in I$, then $u(i) \in F(J,S)$, so $(u(i), u(i)) \in S$. Thus, f and g are mutual inverses, so we have

$$\mathbf{Rel}(\mathrm{Eq}(I), (J,S)) \cong \mathbf{Sets}(I, F(J,S)).$$

We omit the proof of naturality of this bijection and conclude that $\mathrm{Eq} \dashv F$. $\square$

0.2.4. Check that the operation $(I,X) \mapsto \{I \mid X\}$ yields a functor $\mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Sets}$, and show that it is right adjoint to the terminal object functor $\mathbf{Sets} \to \mathrm{Fam}(\mathbf{Sets})$, mapping a set J to the family of singletons $(\{*\})_{j\in J}$. Describe the unit and counit of the adjunction explicitly.

Solution. Define the functor $R : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Sets}$ on objects by

$$R(I,X) = \{I \mid X\} = \sum_{i\in I} X_i = \Big\{(i,x) : i \in I \text{ and } x \in X_i\Big\}$$

and on morphisms by

$$R\left((I,X) \xrightarrow{(u,f)} (J,Y)\right) = \Big((i,x) \longmapsto (u(i), f_i(x))\Big).$$

We claim that $L \dashv R$, where $L : \mathbf{Sets} \to \mathrm{Fam}(\mathbf{Sets})$ is the "terminal object functor," defined on objects by $L(J) = (\{*\})_{j\in J}$. It suffices to prove that there is an isomorphism

$$\mathrm{Fam}(\mathbf{Sets})\left((\{*\})_{j\in J}, (I,X)\right) \cong \mathbf{Sets}\left(J, \{I \mid X\}\right),$$

which is natural in J in **Sets** and (I,X) in $\mathrm{Fam}(\mathbf{Sets})$. Indeed, define

$$\varphi : \mathrm{Fam}(\mathbf{Sets})\left((\{*\})_{j\in J}, (I,X)\right) \longrightarrow \mathbf{Sets}\left(J, \{I \mid X\}\right)$$

by

$$\varphi\left(\left(\{*\}\right)_{j\in J} \xrightarrow{(u,f)} (I,X)\right) = \Big(j \longmapsto (u(j), f_j(*))\Big).$$

One can verify that φ is a bijection, with

$$\varphi^{-1}\left(J \xrightarrow{g} \{I \mid X\}\right) = \left(j \longmapsto \mathrm{fst}(g(j)), \Big(* \longmapsto \mathrm{snd}(g(j))\Big)_{j\in J}\right),$$

and that φ is natural in J and (I,X). Thus, it yields an adjunction $L \dashv R$. The unit of this adjunction is then given by

$$\eta_J = \varphi\left(\mathrm{id}_{(\{*\})_{j\in J}}\right) : J \longrightarrow \sum_{j\in J}\{*\}, \qquad \eta_J(j) = (j,*),$$

and the counit is given by

$$\begin{aligned}
&\varepsilon_{(I,X)} : (\{*\})_{(i,x)\in\{I|X\}} \longrightarrow (I,X)\\
&\varepsilon_{(I,X)} = \varphi^{-1}\left(\mathrm{id}_{\{I|X\}}\right) = \left((i,x) \longmapsto i, \Big(* \longmapsto x\Big)_{(i,x)\in\{I|X\}}\right). \qquad \square
\end{aligned}$$

Chapter 1

Introduction to fibred category theory

1.1 Fibrations

1.1.1.

(i) Prove Proposition 1.1.4.[a]

(ii) Suppose f is Cartesian and g and h are above the same map. Show that $f \circ g = f \circ h$ implies $g = h$.

[a] [7, Proposition 1.1.4] asserts that Cartesian liftings are unique up-to-isomorphism (in a slice): if f and f' with $\operatorname{cod} f = \operatorname{cod} f'$ are both Cartesian over the same map, then there is a unique vertical isomorphism $\varphi : X \xrightarrow{\cong} X'$ with $f' \circ \varphi = f$.

Remark 1.1.1.1. We will prove this in the more general context of displayed categories (cf. [1]). The conclusion of Exercise 1.1.1(i) (i.e., [7, Proposition 1.1.4]) appears in the literature in, e.g., [3, Lemma 8.1.4(2)], [4, Remark 3.3], [5, Lemma 2.3(ii)], and [8, Proposition 9.1.4(4)].

Solution, (i). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ and $f' : X' \to Y$ are Cartesian morphisms in $\mathbb{E}$ over u. Since f' is Cartesian and f is a morphism into Y over $u = u \circ \mathrm{id}_I$, there exists a unique morphism $\varphi : X \to X'$ in $\mathbb{E}$ over I such that $f = f' \circ \varphi$. This is summarized by the following diagram.

$$
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[d] \arrow[dr, bend left, "u"] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
\quad \Longrightarrow \quad
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] \arrow[dr, dashed, "\exists!\,\varphi"] & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
$$

Similarly, there exists a unique morphism $\psi : X' \to X$ in $\mathbb{E}$ over I such that $f' = f \circ \psi$. Moreover, both $\mathrm{id}_{X'}$ and $\varphi \circ \psi$ are morphisms in $\mathbb{E}$ over I whose composition with f' is f' itself, as suggested in the following diagram.

$$\begin{tikzcd}
X' \arrow[dr, "\psi"] \arrow[drr, bend left, "f'"] \arrow[ddr, bend right, "\mathrm{id}_{X'}"'] \arrow[ddd] & & & \\
& X \arrow[dr, "\varphi"] \arrow[drr, bend left, "f"] \arrow[dd] & & \\
I \arrow[dr, equal] & & X' \arrow[r, "f'"] \arrow[dd] & Y \arrow[dd] \\
& I \arrow[dr, equal] & & \\
& & I \arrow[r, "u"] & J
\end{tikzcd}$$

Thus, since f' is Cartesian, $\varphi \circ \psi = \mathrm{id}_{X'}$. By symmetry, $\psi \circ \varphi = \mathrm{id}_X$, and hence φ is an isomorphism. $\square$

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ and $v : K \to I$ are morphisms in $\mathbb{B}$, $f : X \to Y$ is a Cartesian morphism in $\mathbb{E}$ over u, and $g, h : Z \to X$ are morphisms in $\mathbb{E}$ over v such that $f \circ g = f \circ h$. These suppositions are summarized in the following commutative diagram.

$$\begin{tikzcd}
Z \arrow[dr, bend left, "g"] \arrow[drr, bend left, "f\circ g = f\circ h"] \arrow[dr, bend right, "h"'] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Since f is Cartesian, this situation immediately implies that $g = h$. $\square$

1.1.2. Let $p : \mathbb{E} \to \mathbb{B}$ be a functor; assume $f : X \to Y$ is in $\mathbb{E}$ and put $u = pf$. Show that f is Cartesian if and only if for each $Z \in \mathbb{E}$ and $v : pZ \to pX$ in $\mathbb{B}$, the function

$$\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$$

is an isomorphism.

Remark 1.1.2.1. We will prove this in the more general context of displayed categories (cf. [1]). However, this result is essentially an unpacking of the definition of Cartesianness and, in particular, of [1, Definition 5.1].

Solution. Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ is a morphism in $\mathbb{E}$ over u.

($\Longrightarrow$) Suppose f is Cartesian, and take a morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Since f is Cartesian, this situation gives rise to a one-to-one correspondence between the dashed arrows in the diagram below.

$$\begin{tikzcd}
Z \arrow[d] \arrow[dr, dashed] \arrow[drr, dashed, bend left] & & \\
& X \arrow[r, "f"] \arrow[d] \arrow[dr, bend left, "u\circ v"] & Y \arrow[d] \\
K \arrow[r, "v"'] \arrow[urr, bend left] & I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, given a morphism $g : Z \to X$ in $\mathbb{E}$ over v, composition with f yields a morphism $f \circ g : Z \to Y$ over $u \circ v$. Conversely, given a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$, there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $f \circ g = h$ since f is Cartesian. This is precisely the assertion that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism (of sets).

($\Longleftarrow$) Conversely, suppose that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism for every morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Consider a particular morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K, and suppose there is a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Then the assumed isomorphism implies that there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $h = f \circ g$, which shows that f is Cartesian. $\square$

1.1.3. Consider the total category of a fibration. Show that

(i) every morphism factors as a vertical map followed (diagrammatically) by a Cartesian one;

(ii) a Cartesian map above an isomorphism is an isomorphism. Espe-

cially a vertical Cartesian map is an isomorphism.

Solution, (i). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a morphism in $\mathbb{E}$ over a morphism $u : I \to J$ in $\mathbb{B}$. Then there exists a Cartesian morphism $f' : X' \to Y$ in $\mathbb{E}$ above u. Then we are in the situation of the diagram on the left below, so there exists a unique (vertical) morphism $g : X \to X'$ over I such that the diagram below on the right commutes.

$$\begin{tikzcd}
X \arrow[dd] \arrow[drr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[dd] \arrow[ddr, bend left, "u"] & Y \arrow[dd] \\
I \arrow[dr, equal] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
X \arrow[dd] \arrow[dr, dashed, "g"] \arrow[drr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[dd] \arrow[ddr, bend left, "u"] & Y \arrow[dd] \\
I \arrow[dr, equal] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $f = f' \circ g$; in other words, f factors as a vertical morphism followed by a Cartesian morphism. $\square$

Solution, (ii). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a Cartesian morphism in $\mathbb{E}$ over an isomorphism $u : I \to J$ in $\mathbb{B}$. Let $v = u^{-1} : J \to I$ in $\mathbb{B}$, and consider the situation in the following commutative diagram.

$$\begin{tikzcd}
Y \arrow[dd] \arrow[dr, dashed, "\exists! g"] \arrow[drr, bend left, "\mathrm{id}_Y"] & & \\
& X \arrow[r, "f"] \arrow[dd] \arrow[ddr, bend left, "\mathrm{id}_J"] & Y \arrow[dd] \\
J \arrow[dr, "\sim", "v"'] & & \\
& I \arrow[r, "\sim", "u"'] & J
\end{tikzcd}$$

Here the dashed morphism $g : Y \to X$ exists uniquely by virtue of f being Cartesian. Since $f \circ g = \mathrm{id}_Y$ is Cartesian (by Exercise 1.1.4(i)) and f is Cartesian, Exercise 1.1.4(iii) implies that g is Cartesian over the isomorphism v, and $f \circ g = \mathrm{id}_Y$. The same argument then implies that there exists a morphism $h : X \to Y$ over $u = v^{-1}$ such that $g \circ h = \mathrm{id}_X$. But then

$$f = f \circ \mathrm{id}_X = f \circ g \circ h = \mathrm{id}_Y \circ h = h,$$

so $g \circ f = \mathrm{id}_X$, and hence $g = f^{-1}$, so f is an isomorphism. $\square$

1.1.4. Let $\begin{array}{c}\mathbb{E}\\ \downarrow p\\ \mathbb{B}\end{array}$ be a fibration. Prove that

(i) all isomorphisms in $\mathbb{E}$ are Cartesian;

(ii) if $\xrightarrow{f}\xrightarrow{g}$ is a composable pair of Cartesian morphisms in $\mathbb{E}$, then also their composite $\xrightarrow{g\circ f}$ is Cartesian.

Hence it makes sense to talk about the subcategory $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E}$ having all objects from $\mathbb{E}$ but only the Cartesian arrows. We write $|p|$ for the composite $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E} \to \mathbb{B}$.

(iii) Let $\xrightarrow{f}\xrightarrow{g}$ be a composable pair in $\mathbb{E}$ again. Show now that if g and $g \circ f$ are Cartesian, then f is Cartesian as well.

(iv) Verify that a consequence of (iii) is that the functor $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is a fibration. From (ii) in the previous exercise it follows that all fibres of $|p|$ are groupoids (*i.e.* that all maps in the fibres are isomorphisms).

[This $|p|$ will be called the **fibration of objects** of p.]

Remark 1.1.4.1. We will prove this in the more general context of displayed categories (cf. [1]).

Solution, (i). Let $\mathbb{E}$ be a displayed category over $\mathbb{B}$ (not necessarily a fibration), and let $f : X \to Y$ be a displayed isomorphism in $\mathbb{E}$ above an isomorphism $u : I \to J$ in $\mathbb{B}$. Suppose $v : K \to I$ is a morphism in $\mathbb{B}$ and $g : Z \to Y$ is a morphism in $\mathbb{E}$ above $u \circ v : K \to J$. Then $f^{-1} \circ g : Z \to X$ is the unique morphism in $\mathbb{E}$ above v such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "f^{-1}\circ g"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, to verify uniqueness, suppose $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $f \circ h = g$, and observe that

$$h = f^{-1} \circ f \circ h = f^{-1} \circ g.$$

Therefore f is Cartesian. $\square$

Remark 1.1.4.2. The conclusion of Exercise 1.1.4(ii) appears in the literature in, e.g., [3, Lemma 8.1.4(i)]

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$ (not necessarily a fibration), $u : I \to J$ and $v : J \to K$ are morphisms in $\mathbb{B}$, and $f : X \to Y$ and $g : Y \to Z$ are Cartesian morphisms in $\mathbb{E}$ over u and v, respectively. To show that $g \circ f : X \to Z$ is Cartesian over $v \circ u : I \to K$, consider a morphism $w : L \to I$ in $\mathbb{B}$ and a morphism $h : W \to Z$ in $\mathbb{E}$ above $v \circ u \circ w$. These data are summarized in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[rrr, bend left, "h"] \arrow[d] & & & \\
 & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r, "g"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Z \arrow[d] \\
L \arrow[r, "w"] & I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

Since g is Cartesian, there exists a unique morphism $h' : W \to Y$ in $\mathbb{E}$ above $u \circ w$ such that $h = g \circ h'$, as in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[rrr, bend left, "h"] \arrow[rr, bend left, dashed, "h'"] \arrow[d] & & & \\
 & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r, "g"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Z \arrow[d] \\
L \arrow[r, "w"] & I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

Moreover, since f is Cartesian, there exists a unique morphism $h'' : W \to X$

in $\mathbb{E}$ above w such that $h' = f \circ h''$, as in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[dr, dashed, "h''"] \arrow[drr, bend left, "h'"] \arrow[drrr, bend left, "h"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[r, "g"] \arrow[dd] \arrow[dr, phantom, "\lrcorner"] & Z \arrow[dd] \\
L \arrow[dr, "w"] & & & \\
& I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

In particular, we have

$$h = g \circ h' = g \circ f \circ h''.$$

It remains to show that if $i : W \to X$ is another morphism in $\mathbb{E}$ above w such that $h = g \circ f \circ i$, then $i = h''$. Indeed, in this case the uniqueness of h' implies that $f \circ i = h'$, and so the uniqueness of h'' implies that $i = h''$. Thus, $g \circ f$ is Cartesian. $\square$

Solution, (iii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$ (not necessarily a fibration), $u : I \to J$ and $v : J \to K$ are morphisms in $\mathbb{B}$, and $f : X \to Y$ and $g : Y \to Z$ are morphisms in $\mathbb{E}$ over u and v, respectively, such that g and $g \circ f$ are Cartesian. To show that $f : X \to Y$ is Cartesian over u, consider a morphism $w : L \to I$ in $\mathbb{B}$ and a morphism $h : W \to Y$ in $\mathbb{E}$ above $u \circ w$. This situation is summarized in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[drrr, bend left, "h"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[r, "g"] \arrow[dd] & Z \arrow[dd] \\
L \arrow[dr, "w"] & & & \\
& I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

But then $g \circ h : W \to Z$ is a morphism in $\mathbb{E}$ above $v \circ u \circ w$, so since $g \circ f$ is Cartesian, there exists a unique morphism $h' : W \to X$ in $\mathbb{E}$ over w such that $g \circ h = g \circ f \circ h'$. Since g is Cartesian, Exercise 1.1.1(i) implies that $h = f \circ h'$. Moreover, suppose $i : W \to X$ is another morphism above w such that $h = f \circ i$. Then $g \circ h = g \circ f \circ i$, so the uniqueness of h' arising from $g \circ f$ being Cartesian implies that $i = h'$. Thus, f is Cartesian. $\square$

Solution, (iv). Let $p : \mathbb{E} \to \mathbb{B}$ be a fibration; we will show that $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is also a fibration. Consider a morphism $u : I \to J$ in $\mathbb{B}$ and an object Y in $\mathbb{E}$ over J. Since p is a fibration, there exists a p-Cartesian morphism $f : X \to Y$ in $\mathbb{E}$ over u. We claim that f is $|p|$-Cartesian as well. Indeed, suppose there is a morphism $v : K \to I$ in $\mathbb{B}$ and a p-Cartesian morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Since f is p-Cartesian, there exists a unique morphism $h : Z \to X$ in $\mathbb{E}$ over v such that $g = f \circ h$. But now (iii) implies that h is p-Cartesian as well, so the entire diagram below is actually a diagram in the displayed category induced by $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "\exists! h"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend right, "u \circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Therefore, f is $|p|$-Cartesian, so $|p|$ is a fibration. $\square$

1.1.5. Verify that the following two results—known as the Pullback Lemmas—are a consequence of (ii) and (iii) in the previous exercise. Consider

$$\begin{tikzcd}
\bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "(B)"] & \bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "(A)"] & \bullet \arrow[d] \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet
\end{tikzcd}$$

(i) If (A) and (B) are pullback squares, then the outer rectangle is also a pullback square.

(ii) If the outer rectangle and (A) are pullback squares, then (B) is a pullback square as well.

Solution. Recall (e.g., from [4, Example 3.15] or [7, Proposition 1.1.6(ii)])

that in a category $\mathbb{B}$, a commutative square

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "p"] & Y \arrow[d, "q"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

is a pullback square if and only if (u, f) is a Cartesian lift of u in the context of the codomain functor $\mathrm{cod} : \mathbb{B}^{\to} \to \mathbb{B}$, as in the following diagram.

$$\begin{tikzcd} \left(\begin{array}{c} X \\ \downarrow p \\ I \end{array}\right) \arrow[r, "(u,f)"] \arrow[d, "\mathrm{cod}"] & \left(\begin{array}{c} Y \\ \downarrow q \\ J \end{array}\right) \arrow[d, "\mathrm{cod}"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

Therefore, (i) and (ii) follow immediately from Exercise 1.1.4(ii) and Exercise 1.1.4(iii), respectively. □

1.1.6. Consider a functor $p : \mathbb{E} \to \mathbb{B}$. We describe a slightly weaker notion of Cartesianness, than the one above. Call a morphism $f : X \to Y$ in $\mathbb{E}$ **weak Cartesian** if for each $g : Z \to Y$ with $pf = pg$ there is a unique vertical $h : Z \to X$ with $f \circ h = g$. Show that the functor p is a fibration if and only if both

(a) every morphism $u : I \to pY$ in $\mathbb{B}$ has a weak Cartesian lifting $f : X \to Y$;

(b) the composition of two weak Cartesian morphisms is again weak Cartesian.

Remark 1.1.6.1. The conclusion of Exercise 1.1.6 appears in the literature in, e.g., [3, Proposition 8.1.7] and [5, Proposition 2.5].

Solution. ($\Longrightarrow$) Since every Cartesian morphism is weak Cartesian and compositions of Cartesian morphisms are Cartesian by Exercise 1.1.4(ii), every fibration satisfies (a) and (b).

($\Longrightarrow$) Conversely, suppose (a) and (b) hold. Let $u : I \to J$ be a morphism in $\mathbb{B}$, and let Y be an object in $\mathbb{E}$ above J. By (a), let $f : X \to Y$ be a weak Cartesian lift of u in $\mathbb{E}$. Consider another morphism $v : K \to I$ in $\mathbb{B}$ and a morphism $g : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. By (a), choose a weak Cartesian lift $p : W \to X$ in $\mathbb{E}$ of v. Now $f \circ p$ is weak Cartesian over $u \circ v$ by (b), so there exists a unique vertical morphism $q : Z \to W$ over K such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "q"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& W \arrow[r, "p"] \arrow[dd] & X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, equal] & & & \\
& K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, "p\circ q"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"] & & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

It remains to show that if $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $g = f \circ h$, then $h = p \circ q$. In this case, since p is weak Cartesian, we can find a unique vertical morphism $q' : Z \to W$ over K such that $h = p \circ q'$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "q'"] \arrow[drr, bend left, "h"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& W \arrow[r, "p"] \arrow[dd] & X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, equal] & & & \\
& K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $g = f \circ p \circ q'$, so the uniqueness of q implies that $q' = q$, and hence $h = p \circ q$. Thus, f is a Cartesian lift of u, so p is a fibration. $\square$

1.1.7. Check that the following are (trivial) examples of fibrations

$$\begin{tikzcd} \mathbb{B} \times \mathbb{C} \arrow[d, "\mathrm{fst}"] \\ \mathbb{B} \end{tikzcd} \qquad \begin{tikzcd} \mathbb{B} \arrow[d, "\mathrm{id}"] \\ \mathbb{B} \end{tikzcd} \qquad \begin{tikzcd} \mathbb{B} \arrow[d] \\ 1 = \{*\} \end{tikzcd} \qquad \begin{tikzcd} X \arrow[d] \\ I \end{tikzcd}$$

where X, I are sets (*i.e.* discrete categories).

Solution. Consider the functor $\mathbb{B} \times \mathbb{C} \xrightarrow{\mathrm{fst}} \mathbb{B}$, which projects onto the first component. Consider a morphism $u : I \to J$ in $\mathbb{B}$ and an object (J, Y) in $\mathbb{B} \times \mathbb{C}$ above J. We claim that $(v, \mathrm{id}_Y) : (I, Y) \to (J, Y)$ is a Cartesian lift of u. Suppose there are morphisms $v : K \to J$ in $\mathbb{B}$ and $(w, g) : (K, Z) \to (J, Y)$ in $\mathbb{B} \times \mathbb{C}$ above $u \circ v$ (i.e., $w = u \circ v$), as in the following diagram.

$$\begin{tikzcd} (K,Z) \arrow[dr, dashed, "?"] \arrow[drr, bend left, "(w,g)"] \arrow[dd, "\mathrm{fst}"] & & \\ & (I,Y) \arrow[r, "(u,\mathrm{id}_Y)"] \arrow[d, "\mathrm{fst}"] & (J,Y) \arrow[d, "\mathrm{fst}"] \\ K \arrow[r, "v"] & I \arrow[r, "u"] & J \end{tikzcd}$$

Observe that $(v, g) : (K, Z) \to (I, Y)$ is the unique morphism in $\mathbb{B} \times \mathbb{C}$ that fills in the dashed "?" morphism making the diagram above commute, so (u, id_Y) is Cartesian and hence $\mathbb{B} \times \mathbb{C} \xrightarrow{\mathrm{fst}} \mathbb{B}$ is a fibration.

Next, consider the identity functor $\mathbb{B} \xrightarrow{\mathrm{id}} \mathbb{B}$, and observe that this situation is equivalent to the functor $\mathbb{B} \times 1 \xrightarrow{\mathrm{fst}} \mathbb{B}$, so it is a fibration by the first part of this exercise. More explicitly, consider a morphism $u : I \to J$ in $\mathbb{B}$. The only object above J in this scenario is J itself, and we claim that u itself is the unique Carteisan lift of u. Indeed, given morphisms $v : K \to I$ in $\mathbb{B}$ ("below") and $w : K \to J$ in $\mathbb{B}$ ("above") over $u \circ v$ (i.e., $w = u \circ v$), the

unique morphism $K \to I$ which fills in the dashed "?" arrow in the diagram below is v itself.

$$\begin{tikzcd} K \arrow[rr, bend left, "w"] \arrow[dr, dashed, bend left, "?"] \arrow[dr, bend right, "v"'] & & J \\ & I \arrow[ur, "u"] & \end{tikzcd}$$

Thus, u is a Cartesian lift of u, and $\mathbb{B} \xrightarrow{\mathrm{id}} \mathbb{B}$ is a fibration.

Consider the functor $\mathbb{B} \to 1$, and observe that this situation is equivalent to the functor $1 \times \mathbb{B} \xrightarrow{\mathrm{fst}} 1$, so it is a fibration by the first part of this exercise. More explicitly, consider *the* morphism $* \to *$ in 1, and let Y be an object in $\mathbb{B}$ (vacuously above $*$). Then $\mathrm{id}_Y : Y \to Y$ is a Cartesian lift of $* \to *$, whence $\mathbb{B} \to 1$ is a fibration.

Finally, let X and Y be sets and $p : X \to I$ a function, considered as a functor between discrete categories. All morphisms in I and X are identities, so if $i \in I$ and $x \in X$ is a "lift" of i (i.e., $p(x) = i$ or equivalently $x \in p^{-1}(i)$), then id_x is a Cartesian lift of id_i. Therefore, $p : X \to I$ is a fibration. $\square$

1.1.8. For an arbitrary category $\mathbb{B}$, consider the domain functor $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$.

(i) Describe the fibre category above $I \in \mathbb{B}$. It is usually called the **opslice category** or simply **opslice** and written as $I \setminus \mathbb{B}$.

(ii) Show that dom is a fibration (without any assumptions about $\mathbb{B}$).

(iii) Show also that for each $I \in \mathbb{B}$ the domain functor $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$ is a fibration.

Solution, (i). For a category $\mathbb{B}$ and an object $I \in \mathbb{B}$, the category $I \setminus \mathbb{B}$ has

objects Morphisms $\begin{pmatrix} I \\ \downarrow\varphi \\ X \end{pmatrix}$ in $\mathbb{B}$ with domain I.

morphisms Given objects $\begin{pmatrix} I \\ \downarrow\varphi \\ X \end{pmatrix}$ and $\begin{pmatrix} I \\ \downarrow\psi \\ Y \end{pmatrix}$ in $I \setminus \mathbb{B}$, a morphism in $I \setminus \mathbb{B}$ from the first to the second consists of a morphism $f : X \to Y$ in $\mathbb{B}$

such that the following diagram commutes.

$$\begin{tikzcd} & I \arrow[dl, "\varphi"'] \arrow[dr, "\psi"] & \\ X \arrow[rr, "f"] & & Y \end{tikzcd}$$

□

Solution, (ii). Consider a morphism $u : I \to J$ in $\mathbb{B}$, and another morphism $\psi : J \to Y$ in $\mathbb{B}$ (i.e., an object $\begin{pmatrix} J \\ \downarrow\psi \\ Y \end{pmatrix}$ in $\mathbb{B}^{\to}$ over J with respect to $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$). We claim that

$$\begin{pmatrix} I \\ \downarrow \psi \circ u \\ Y \end{pmatrix} \xrightarrow{(\mathrm{id}_Y, u)} \begin{pmatrix} J \\ \downarrow \psi \\ Y \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd} I \arrow[r, "u"] \arrow[d, "\psi\circ u"'] & J \arrow[d, "\psi"] \\ Y \arrow[r, equal] & Y \end{tikzcd} \tag{1.1.8.1}$$

is a Cartesian lift of u. Indeed, consider a situation as in the following commutative diagram of solid arrows.

$$\begin{tikzcd} \begin{pmatrix} K \\ \downarrow \varphi \\ Z \end{pmatrix} \arrow[rr, bend left, "(g, u\circ v)"] \arrow[dr, dashed, "?"] \arrow[dd, "\mathrm{dom}"'] & & \\ & \begin{pmatrix} I \\ \downarrow \psi \circ u \\ Y \end{pmatrix} \arrow[r, "(\mathrm{id}_Y, u)"] \arrow[dd, "\mathrm{dom}"] & \begin{pmatrix} J \\ \downarrow \psi \\ Y \end{pmatrix} \arrow[dd, "\mathrm{dom}"] \\ K \arrow[dr, "v"'] & & \\ & I \arrow[r, "u"] & J \end{tikzcd} \tag{1.1.8.2}$$

We must find a unique morphism in $\mathbb{B}^{\to}$ to fill in the dashed "?" arrow in (1.1.8.2) that makes the diagram commute. But observe that (1.1.8.2) is

equivalent to the following diagram.

$$\begin{tikzcd}
K \arrow[r, "v"] \arrow[d, "\varphi"] & I \arrow[r, "u"] \arrow[d, "\psi\circ u"] & J \arrow[d, "\psi"] \\
Z \arrow[r, dashed, "?"] \arrow[rr, bend right, "g"'] & Y \arrow[r, equal] & Y
\end{tikzcd}$$

It becomes evident that the unique morphism that fills in the dashed "?" arrow in (1.1.8.2) to makes the diagram commute is $? = (g, v)$. Thus, (1.1.8.1) is a Cartesian lift of u, and hence $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$ is a fibration. $\square$

Solution, (iii). The same proof as in (ii) applies here as well. Explicitly, fix an object I in $\mathbb{B}$, let $u : X \to Y$ be a morphism in $\mathbb{B}$, and let $\begin{pmatrix} Y \\ \downarrow\psi \\ I \end{pmatrix}$ in $\mathbb{B}/I$ over Y relative to the functor $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$. Then a straightforward modification of the proof of (ii) shows that

$$\begin{pmatrix} X \\ \downarrow \psi\circ u \\ I \end{pmatrix} \xrightarrow{u} \begin{pmatrix} Y \\ \downarrow \psi \\ I \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd}
X \arrow[rr, "u"] \arrow[dr, "\psi\circ u"'] & & J \arrow[dl, "\psi"] \\
& I &
\end{tikzcd}$$

is a Cartesian lift of u, and hence $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$ is a fibration. $\square$

1.1.9. Assume $\mathbb{B}$ is a category with pullbacks. Show that the functor $\mathbb{B}^{\to\to} \to \mathbb{B}^{\to}$ sending $\xrightarrow{f}\xrightarrow{g}$ to $\xrightarrow{g}$ is a fibration. Is the composite $B^{\to\to} \to \mathbb{B}^{\to} \to \mathbb{B}$ also a fibration?

Solution. Let

$$\begin{pmatrix} X \\ \downarrow \varphi \\ I \end{pmatrix} \xrightarrow{(u,f)} \begin{pmatrix} Y \\ \downarrow \psi \\ J \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "\varphi"'] & Y \arrow[d, "\psi"] \\
I \arrow[r, "u"] & J
\end{tikzcd}$$

be a morphism in $\mathbb{B}^{\to}$, and let $\begin{pmatrix} B \\ \downarrow\beta \\ Y \\ \downarrow\psi \\ I \end{pmatrix}$ be an object in $\mathbb{B}^{\to\to}$ "above" $\begin{pmatrix} Y \\ \downarrow\psi \\ I \end{pmatrix}$.

Since $\mathbb{B}$ has pullbacks, let $X \xleftarrow{\alpha} A \xrightarrow{q} B$ be a pullback of $X \xrightarrow{f} Y \xleftarrow{\beta} B$. We claim that

$$\begin{pmatrix} A \\ \downarrow\alpha \\ X \\ \downarrow\varphi \\ I \end{pmatrix} \xrightarrow{(u,f,q)} \begin{pmatrix} B \\ \downarrow\beta \\ Y \\ \downarrow\psi \\ J \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd} A \arrow[r, "q"] \arrow[d, "\alpha"'] \arrow[dr, phantom, "\lrcorner"] & B \arrow[d, "\beta"] \\ X \arrow[r, "f"] \arrow[d, "\varphi"'] & Y \arrow[d, "\psi"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

is a Cartesian lift of (u, f). Indeed, consider a situation as in the following diagram.

$$\begin{tikzcd} \begin{pmatrix} C \\ \downarrow\gamma \\ Z \\ \downarrow\xi \\ K \end{pmatrix} \arrow[dr, dashed, "?"] \arrow[drr, bend left, "(w,h,r)"] \arrow[dd] & & \\ & \begin{pmatrix} A \\ \downarrow\alpha \\ X \\ \downarrow\varphi \\ I \end{pmatrix} \arrow[r, "(u,f,q)"] \arrow[dd] & \begin{pmatrix} B \\ \downarrow\beta \\ Y \\ \downarrow\psi \\ J \end{pmatrix} \arrow[dd] \\ \begin{pmatrix} Z \\ \downarrow\xi \\ K \end{pmatrix} \arrow[dr, "(v,g)"'] & & \\ & \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix} \arrow[r, "(u,f)"] & \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix} \end{tikzcd} \tag{1.1.9.1}$$

(where necessarily $w = u \circ v$ and $h = f \circ g$). We must find a unique filler for the dashed morphism labeled "?"; note that its first and second components

are necessarily v and g, respectively. Also note that the diagram above is equivalent to the following one.

$$\begin{tikzcd}
C \arrow[r, dashed, "p"] \arrow[rr, bend left, "r"] \arrow[d, "\gamma"] & A \arrow[r, "q"] \arrow[d, "\alpha"] \arrow[dr, phantom, "\lrcorner"] & B \arrow[d, "\beta"] \\
Z \arrow[r, "g"] \arrow[rr, bend right, "h"'] \arrow[d, "\xi"] & X \arrow[r, "f"] \arrow[d, "\varphi"] & Y \arrow[d, "\psi"] \\
K \arrow[r, "v"] \arrow[rr, bend right, "w"'] & I \arrow[r, "u"] & J
\end{tikzcd}$$

Here the dashed arrow $p : C \to A$ is the unique morphism in $\mathbb{B}$ such that $g \circ \gamma = \alpha \circ p$ and $r = q \circ p$, by virtue of $X \xleftarrow{\alpha} A \xrightarrow{q} B$ being a pullback of $X \xrightarrow{f} Y \xleftarrow{\beta} B$. It follows that the unique filler for "?" in (1.1.9.1) is $? = (v, g, p)$, so (u, f, q) is a Cartesian lift of (u, f), and hence the functor $\mathbb{B}^{\to\to} \to \mathbb{B}^{\to}$ sending $\xrightarrow{f}\xrightarrow{g}$ to $\xrightarrow{g}$ is a fibration.

Moreover, the composite $B^{\to\to} \to \mathbb{B}^{\to} \to \mathbb{B}$ is also a fibration since the composition of two fibrations is again a fibration (cf. [3, Proposition 8.1.12], [4, Proposition 3.7], [5, Proposition 3.1] [8, Proposition 9.1.10]) and since the codomain functor $\mathrm{cod} : \mathbb{B}^{\to} \to \mathbb{B}$ is a fibration when $\mathbb{B}$ has pullbacks (cf. [7, Proposition 1.1.6]). $\square$

1.1.10. Let **Fld** be the category of fields and field homomorphisms; **Vect** is the category of vector spaces: objects are triples $(K, V, \cdot)$ where K is a field of scalars, V is an Abelian group of vectors and $\cdot : K \times V \to V$ is an action of scalar multiplication (which distributes both over scalar and over vector addition). A morphism $(K, V, \cdot) \to (L, W, \cdot)$ in **Vect** is a pair (u, f) where $u : K \to L$ is a field homomorphism and $f : V \to W$ a group homomorphism such that $f(a \cdot x) = u(a) \cdot f(x)$ for all $a \in K$ and $x \in V$.

Check that the obvious forgetful functor **Vect** $\to$ **Fld** is a fibration. What are the fibres? Which maps are Cartesian?

Solution. First, the fibres of **Vect** $\to$ **Fld** are the categories of vector spaces over a fixed field K, with morphisms being ordinary K-linear maps.

Next, we show that **Vect** $\to$ **Fld** is a fibration. Let $u : K \to L$ be a field homomorphism, and let $(L, V, \cdot)$ be an object of **Vect** over L. Then we claim that (u, id_V) is a Cartesian lift of u. Indeed, suppose $v : F \to K$ is another field homomorphism and $(w, g) : (F, W, \cdot) \to (L, V, \cdot)$ be a morphism in **Vect** over $u \circ v$ such that the following diagram commutes.

$$\begin{tikzcd}
(F, W, \cdot) \arrow[dr, dashed, "?"] \arrow[drr, bend left, "(w,g)"] \arrow[dd] & & \\
& (K, V, \cdot) \arrow[r, "(u,\mathrm{id}_V)"] \arrow[d] & (L, V, \cdot) \arrow[d] \\
F \arrow[r, "v"] & K \arrow[r, "u"] & L
\end{tikzcd}$$

The unique filler of the dashed "?" morphism in this diagram is $? = (v, g)$, so (u, id_V) is Cartesian, and hence **Vect** $\to$ **Fld** is Cartesian.

Finally, we claim that a morphism $(u, f) : (K, V, \cdot) \to (L, W, \cdot)$ is Cartesian if and only if $f : V \to W$ is an abelian group isomorphism. To show that abelian group isomorphisms induce Cartesian morphisms, mimic the proof above that showed that (u, id_V) is a Cartesian lift of $u : K \to L$ whenever $(L, V, \cdot)$ is an object in **Vect** above L. Conversely, suppose $(u, f) : (K, V, \cdot) \to (L, W, \cdot)$ is a Cartesian morphism in **Vect** above a field homomorphism $u : K \to L$. We claim that $f : V \to W$ is an abelian group isomorphism. We already showed that in this case $(u, \mathrm{id}_W) : (K, W, \cdot) \to (L, W, \cdot)$ is a Cartesian lift of u, so by Exercise 1.1.1(i), there exists a unique isomorphism $(w, g) : (K, V, \cdot) \to (K, W, \cdot)$ in **Vect** such that $(u, f) = (u, \mathrm{id}_W) \circ (w, g)$. In particular, g is an abelian group isomorphism and $f = \mathrm{id}_W \circ g = g$, so f is an abelian group isomorphism. $\square$

1.2 Some concrete examples: sets, ω-sets and PERs

1.2.1. Prove that a morphism $\left(u, (f_i)_{i \in I}\right)$ in $\mathrm{Fam}(\mathbb{C})$ is Cartesian if and only if each f_i is an isomorphism in $\mathbb{C}$.

Solution. ($\Longrightarrow$) Suppose

$$\left(u, (f_i)_{i\in I}\right) : (X_i)_{i\in I} \longrightarrow (Y_j)_{j\in J}$$

is a Cartesian morphism in $\mathrm{Fam}(\mathbb{C})$ over a morphism $u : I \to J$ in $\mathbb{C}$. We also know that

$$\left(u, \left(\mathrm{id}_{Y_{u(i)}}\right)_{i\in I}\right) : \left(Y_{u(i)}\right)_{i\in I} \longrightarrow (Y_j)_{j\in J}$$

is another Cartesian lift of u (cf. [7, preceding Definition 1.2.1]). Thus, by Exercise 1.1.1(i), there is a unique vertical isomorphism

$$\left(\mathrm{id}_I, \left(X_i \xrightarrow{g_i} Y_{u(i)}\right)_{i\in I}\right) : (X_i)_{i\in I} \longrightarrow \left(Y_{u(i)}\right)_{i\in I}$$

such that

$$\left(u, (f_i)_{i\in I}\right) = \left(u, \left(\mathrm{id}_{Y_{u(i)}}\right)_{i\in I}\right) \circ \left(\mathrm{id}_I, (g_i)_{i\in I}\right)$$

in $\mathrm{Fam}(\mathbb{C})$. In particular, $f_i = \mathrm{id}_{Y_{u(i)}} \circ g_i = g_i$ for all $i \in I$, so each f_i is an isomorphism.

($\Longleftarrow$) Conversely, suppose

$$\left(u, (f_i)_{i\in I}\right) : (X_i)_{i\in I} \longrightarrow (Y_j)_{j\in J}$$

is a morphism in $\mathrm{Fam}(\mathbb{C})$ over a morphism $u : I \to J$ in $\mathbb{C}$ such that each $f_i : X_i \to Y_{u(i)}$ is an isomorphism. We claim that $\left(u, (f_i)_{i\in I}\right)$ is Cartesian. Suppose $v : K \to I$ is another morphism in $\mathbb{C}$ and

$$\left(u \circ v, (g_k)_{k\in K}\right) : (Z_k)_{k\in K} \longrightarrow (Y_j)_{j\in J}$$

is a morphism in $\mathrm{Fam}(\mathbb{C})$ over $u \circ v$, as in the following diagram.

$$\begin{tikzcd}
(Z_k)_{k\in K} \arrow[dr, dashed, "?"] \arrow[drr, bend left, "\left(u\circ v, (g_k)_{k\in K}\right)"] \arrow[dd] & & \\
& (X_i)_{i\in I} \arrow[r, "\left(u, (f_i)_{i\in I}\right)"] \arrow[d] & (Y_j)_{j\in J} \arrow[d] \\
K \arrow[dr, "v"'] & I \arrow[r, "u"] & J \\
& &
\end{tikzcd}$$

Then the unique filler of the dashed "?" morphism in this diagram is $? = \left(v, \left(f_{v(k)}^{-1} \circ g_k\right)_{k\in K}\right)$, so $\left(u, (f_i)_{i\in I}\right)$ is Cartesian. $\square$

1.2.2. Consider a map $f = (f_i : X_i \to Y_i)_{i \in I}$ in the fibre $\mathrm{Fam}(\mathbb{C})_I = \mathbb{C}^I$ over $I \in \mathbf{Sets}$. Prove that f is a mono in this fibre if and only if each f_i is a mono in $\mathbb{C}$.

Solution. ($\Longrightarrow$) Suppose $f = (f_i : X_i \to Y_i)_{i \in I}$ is monic in $\mathbb{C}^I$, and fix $i \in I$. Suppose $g, h : Z \to X_i$ are arrows in $\mathbb{C}$ such that $f_i \circ g = f_i \circ h$. We claim that $h = g$, which will show that f_i is monic. Define a family of object $(Z_j)_{j \in I}$ in $\mathbb{C}^I$, where

$$Z_j = \begin{cases} Z & \text{if } i = j \\ X_j & \text{otherwise,} \end{cases}$$

and define morphisms $g' = \left(g'_j : Z_j \to X_j\right)_{j \in I} : (Z_j)_{j \in I} \to (X_j)_{j \in I}$ and $h' = \left(h'_j : Z_j \to X_j\right)_{j \in I} : (Z_j)_{j \in I} \to (X_j)_{j \in I}$ in $\mathbb{C}^I$, where

$$g'_j = \begin{cases} g & \text{if } i = j \\ \mathrm{id}_{X_j} & \text{otherwise,} \end{cases} \qquad h'_j = \begin{cases} h & \text{if } i = j \\ \mathrm{id}_{X_j} & \text{otherwise,} \end{cases}$$

Then we have $f \circ g' = f \circ h'$, so $g' = h'$, and in particular $g = g'_i = h'_i = h$.

($\Longleftarrow$) Suppose $f = (f_i : X_i \to Y_i)_{i \in I}$ is a morphism in $\mathbb{C}^I$ and f_i is monic in $\mathbb{C}$ for each $i \in I$. Consider morphisms $g = (g_j : Z_j \to X_j)_{j \in I} : (Z_j)_{j \in I} \to (X_j)_{j \in I}$ and $h = (h_j : Z_j \to X_j)_{j \in I} : (Z_j)_{j \in I} \to (X_j)_{j \in I}$ in $\mathbb{C}^I$ such that $f \circ g = f \circ h$. In particular, $f_i \circ g_i = f_i \circ h_i$ for each $i \in I$, so $g_i = h_i$ (since f_i is monic) for each $i \in I$. Therefore, $g = h$, and hence f is monic. $\square$

1.2.3. For an arbitrary category $\mathbb{B}$, let $\mathbb{B}^{\to}_{\bullet}$ be the category with **pointed families** as objects; these are pairs $\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle$ where s is a section of φ (*i.e.* a map $s : I \to X$ with $\varphi \circ s = \mathrm{id}$). A morphism $\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle \to \left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle$ in $\mathbb{B}^{\to}_{\bullet}$ consists of a pair of morphisms $u : I \to J$, $f : X \to Y$ in $\mathbb{B}$ with $\psi \circ f = u \circ \varphi$ and also $f \circ s = t \circ u$. Thus morphisms of pointed

families preserve the points (*i.e.* sections) of the families. Prove that

(i) if the category $\mathbb{B}$ has pullbacks then the functor $\mathbb{B}^{\to}_{\bullet}$ sending $\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle$ to the index set I is a fibration;

(ii) for $\mathbb{B} = \mathbf{Sets}$, there is an equivalence of categories $\mathrm{Fam}(\mathbf{Sets}_\bullet) \xrightarrow{\simeq} \mathbf{Sets}^{\to}_{\bullet}$ like in Proposition 1.2.2, where $\mathbf{Sets}_\bullet$ is the category of **pointed sets**: objects are sets containing a distinguished base point, morphisms are functions preserving such points.

Solution, (i). Let $\mathbb{B}$ be a category with pullbacks, and consider the functor

$$\begin{tikzcd}
\mathbb{B}^{\to}_{\bullet} \arrow[r] & \mathbb{B} \\
\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle \arrow[r, mapsto] \arrow[d, "{(u,f)}"', mapsto] & I \arrow[d, "u"] \\
\left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle \arrow[r, mapsto] & J
\end{tikzcd}$$

where the morphism (u, f) represents the commutative diagram

$$\begin{tikzcd}
I \arrow[r, "u"] \arrow[d, "s"] \arrow[dd, bend right=60, "\mathrm{id}_I"'] & J \arrow[d, "t"'] \arrow[dd, bend left=60, "\mathrm{id}_J"] \\
X \arrow[r, "f"] \arrow[d, "\varphi"] & Y \arrow[d, "\psi"'] \\
I \arrow[r, "u"] & J
\end{tikzcd} \tag{1.2.3.1}$$

We claim that $\mathbb{B}^{\to}_{\bullet} \to \mathbb{B}$ is a fibration. To this end, let $u : I \to J$ be a

morphism in $\mathbb{B}$, and let $\left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle$ be an object in $\mathbb{B}^{\to}_{\bullet}$ above J. Since $\mathbb{B}$ has pullbacks, let $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ be a pullback of $I \xrightarrow{u} J \xleftarrow{\psi} Y$. Moreover, define $s : I \to X$ as the unique dashed arrow that makes the following diagram commute, by virtue of $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ being a pullback of $I \xrightarrow{u} J \xleftarrow{\psi} Y$.

$$\begin{tikzcd}
I \arrow[r, "u"] \arrow[dr, dashed, "s"] \arrow[ddr, bend right, "\mathrm{id}_I"'] & J \arrow[dr, "t"] & \\
& X \arrow[r, "f"] \arrow[d, "\varphi"] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, "\psi"] \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $\varphi \circ s = \mathrm{id}_I$, so $\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle$ is an object in $\mathbb{B}^{\to}_{\bullet}$. Also, we have a situation as in (1.2.3.1), namely that

$$\psi \circ f = u \circ \varphi, \qquad\qquad f \circ s = t \circ u,$$

so we have a morphism

$$\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle \xrightarrow{(u,f)} \left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle \tag{1.2.3.2}$$

in $\mathbb{B}^{\to}_{\bullet}$. We claim that it is a Cartesian lift of u. Indeed, consider a morphism $v : K \to I$ in $\mathbb{B}$ and a morphism

$$\left\langle \begin{pmatrix} Z \\ \downarrow\xi \\ K \end{pmatrix}, r \right\rangle \xrightarrow{(u\circ v,g)} \left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle$$

in $\mathbb{B}^{\to}_{\bullet}$ above $u \circ v$. These data are summarized by the solid arrows in the

following commutative diagram.

$$\begin{tikzcd}
\left\langle \begin{pmatrix} Z \\ \downarrow \xi \\ K \end{pmatrix}, r \right\rangle \arrow[rr, bend left, "(u\circ v, g)"] \arrow[r, dashed, "?"] \arrow[d] & \left\langle \begin{pmatrix} X \\ \downarrow \varphi \\ I \end{pmatrix}, s \right\rangle \arrow[r, "(u,f)"] \arrow[d] & \left\langle \begin{pmatrix} Y \\ \downarrow \psi \\ J \end{pmatrix}, t \right\rangle \arrow[d] \\
K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd} \tag{1.2.3.3}$$

We must find a unique filler for the dashed arrow above v labeled "?" in (1.2.3.3) that makes the diagram commute. Moreover, the situation in (1.2.3.3) can also be summarized by the following commutative diagram in $\mathbb{B}$.

$$\begin{tikzcd}
K \arrow[r, "v"] \arrow[d, "r"] \arrow[dd, bend right=60, "\mathrm{id}_K"'] & I \arrow[r, "u"] \arrow[d, "s"] \arrow[dd, bend right=60, "\mathrm{id}_I"'] & J \arrow[d, "t"'] \arrow[dd, bend left=60, "\mathrm{id}_J"] \\
Z \arrow[d, "\xi"] \arrow[urr, bend left] & X \arrow[r, "f"] \arrow[d, "\varphi"] \arrow[urr, bend left, "g"] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d, "\psi"] \\
K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd} \tag{1.2.3.4}$$

In particular, since $\psi \circ g = u \circ v \circ \xi$, the universal property of the pullback $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ implies that there exists a unique morphism $h : Z \to X$ in $\mathbb{B}$ such that the following diagram in $\mathbb{B}$ commutes.

$$\begin{tikzcd}
Z \arrow[d, "\xi"'] \arrow[dr, dashed, "h"] \arrow[drr, bend left, "g"] & & \\
K \arrow[dr, "v"'] & X \arrow[r, "f"] \arrow[d, "\varphi"'] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d, "\psi"] \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Also, since we have both

$$\begin{aligned}&\psi \circ f \circ s \circ v = u \circ \varphi \circ s \circ v,\\ &\psi \circ f \circ h \circ r = \psi \circ g \circ r = u \circ v \circ \xi \circ r = u \circ \varphi \circ h \circ r,\end{aligned}$$

the universal property of the pullback $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ implies that $s \circ v = h \circ r$, as in the following commutative diagram in $\mathbb{B}$.

$$\begin{tikzcd}
K \arrow[dr, bend left, "s\circ v"] \arrow[dr, bend right, "h\circ r"'] \arrow[dr, phantom, "/\!/"] & & \\
& X \arrow[r, "f"] \arrow[d, "\varphi"'] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d, "\psi"] \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Thus, (1.2.3.4) can be extended into the following commutative diagram.

$$\begin{tikzcd}
K \arrow[r, "v"] \arrow[d, "r"] \arrow[dd, bend right=60, "\mathrm{id}_K"'] & I \arrow[r, "u"] \arrow[d, "s"] \arrow[dd, bend right=60, "\mathrm{id}_I"'] & J \arrow[d, "t"'] \arrow[dd, bend left=60, "\mathrm{id}_J"] \\
Z \arrow[r, "h"'] \arrow[d, "\xi"] \arrow[ur, bend left] & X \arrow[r, "f"] \arrow[d, "\varphi"] \arrow[ur, bend left, "g"] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d, "\psi"] \\
K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

It follows that one choice of "?" filler in (1.2.3.3) is $? = (v, h)$, and its uniqueness follows from the uniqueness of h since $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ is a pullback. Thus, (1.2.3.2) is Cartesian, and hence $\mathbb{B}^{\to}_{\bullet} \to \mathbb{B}$ is a fibration. $\square$

Solution, (ii). We define functors

$$F : \mathrm{Fam}(\mathbf{Sets}_\bullet) \longrightarrow \mathbf{Sets}^{\to}_{\bullet}, \qquad G : \mathbf{Sets}^{\to}_{\bullet} \longrightarrow \mathrm{Fam}(\mathbf{Sets}_\bullet)$$

on objects by

$$F\left((X_i, x_i)_{i\in I}\right) = \left\langle \begin{pmatrix} \sum_{i\in I} X_i \\ \downarrow_{\mathrm{fst}} \\ I \end{pmatrix}, i \longmapsto (i, x_i) \right\rangle$$

and

$$G\left(\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle\right) = \left(\varphi^{-1}(\{i\}), s(i)\right)_{i\in I}$$

and on morphisms by

$$F\left((X_i, x_i)_{i\in I} \xrightarrow{(u,f)} (Y_j, y_j)_{j\in J}\right) = \left(u, (i,x) \longmapsto (u(i), f_i(x))\right)$$

and

$$G\left(\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle \xrightarrow{(u,f)} \left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle\right) = \left(u, \left(f|_{\varphi^{-1}(\{i\})}\right)_{i\in I}\right).$$

Then we have

$$G\left(F\left((X_i, x_i)_{i\in I}\right)\right) = \left(\{i\} \times X_i, x_i\right)_{i\in I} \cong (X_i, x_i)_{i\in I}$$

in $\mathrm{Fam}(\mathbf{Sets}_\bullet)$ and

$$F\left(G\left(\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle\right)\right) = \left\langle \begin{pmatrix} \sum_{i\in I} \varphi^{-1}(\{i\}) \\ \downarrow\mathrm{fst} \\ I \end{pmatrix}, i \longmapsto (i, s(i)) \right\rangle$$
$$\cong \left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle$$

in $\mathbf{Sets}_\bullet^{\to}$. We omit the proofs of the naturality of these isomorphisms, and conclude that $G \circ F \cong \mathrm{id}_{\mathrm{Fam}(\mathbf{Sets}_\bullet)}$ and $F \circ G \cong \mathrm{id}_{\mathbf{Sets}_\bullet^{\to}}$, and that, in particular, there is an equivalence of categories $F : \mathrm{Fam}(\mathbf{Sets}_\bullet) \xrightarrow{\simeq} \mathbf{Sets}_\bullet^{\to}$ □

1.2.4. Check that for a PER R one has $R \subseteq |R|^2$.

Solution. Let R be a partial equivalence relation (on an arbitrary set, not necessarily just $\mathbb{N}$), and let $(x, y) \in R$. Since R is symmetric, we also have $(y, x) \in R$, and since R is transitive, it follows that $(x, x) \in R$ and $(y, y) \in R$. Thus, $x \in |R|$ and $y \in |R|$, so $(x, y) \in |R|^2$. Therefore $R \subseteq |R|^2$. □

1.2.5. Let I be a set. A **partition** of I is a collection $Q \subseteq P(I)$ of subsets of I satisfying (1) every set in Q is non-empty (2) if $a, b \in Q$ and $a \cap b \neq \emptyset$, then $a = b$ (3) $\bigcup Q = I$. A **partial partition** of I is a subset $Q \subseteq P(I)$ satisfying (1) and (2) but not necessarily (3). Show that

(i) there is a bijective correspondence between partitions and equivalence relations and between partial partitions and partial equivalence relations (on I);

(ii) there is a bijective correspondence between partial equivalence relations on I and equivalence relations on subsets of I.

Solution, (i). For a set I, let $\mathrm{PPart}(I)$ and $\mathrm{PER}(I)$ denote the sets of partial partitions on I and partial equivalence relations on I, respectively. Within these, let $\mathrm{Part}(I) \subseteq \mathrm{PPart}(I)$ and $\mathrm{ER}(I) \subseteq \mathrm{PER}(I)$ denote the sets of (total) partitions on I and (total) equivalence relations on I, respectively. Define

$$f : \mathrm{PPart}(I) \longrightarrow \mathrm{PER}(I), \qquad g : \mathrm{PER}(I) \longrightarrow \mathrm{PPart}(I)$$

by

$$\begin{aligned} f(Q) &= \Big\{(x, y) \in I^2 : \text{there exists } a \in Q \text{ such that } x, y \in a\Big\}, \\ g(R) &= \Big\{\text{equivalence classes of } R\Big\}. \end{aligned}$$

A routine verification shows that f and g are mutual inverses, and furthermore they restrict to mutually inverse bijections between $\mathrm{Part}(I)$ and $\mathrm{ER}(I)$. $\square$

Solution, (ii). Note that for distinct subsets J and K of I, the sets $\mathrm{ER}(J)$ and $\mathrm{ER}(K)$ are disjoint, so we have

$$\mathrm{PER}(I) = \bigcup_{J \subseteq I} \mathrm{ER}(J) \cong \coprod_{J \subseteq I} \mathrm{ER}(J). \qquad \square$$

1.2.6. Notice that for $R \in \mathbf{PER}$, the "global sections" or "global elements" homset $\mathbf{PER}(1, R)$ is isomorphic to the quotient $\mathbb{N}/R$. And also that all homsets in $\mathbf{PER}$ and in ω-$\mathbf{Sets}$ are countable.

Proof. First, fix $R \in \mathbf{PER}$. For a terminal object 1 in $\mathbf{PER}$, choose $1 = \{(0,0)\}$, and denote $\mathbb{N}/1 = \{[0]_1\} = *$. Define $\varphi : \mathbf{PER}(1, R) \to \mathbb{N}/R$ by

$$\varphi\left(* \xrightarrow{f} \mathbb{N}/R\right) = f([0]_1).$$

Then φ is injective (since $[0]_1$ is the only element of $*$). Moreover, take an equivalence class $[n]_R \in \mathbb{N}/R$ for some $n \in |R|$. Let $e \in \mathbb{N}$ be the code of the constant total recursive function $\mathbb{N} \to \mathbb{N}$, $_ \mapsto n$. Then define $f : * \to \mathbb{N}/R$ by $f([0]_1) = [n]_R$, and note that f is tracked by e, so $f \in \mathbf{PER}(1, R)$. Then we have $\varphi(f) = [n]_R$, so φ is surjective. Thus, φ is an isomorphism (in $\mathbf{Sets}$) between $\mathbf{PER}(1, R)$ and $\mathbb{N}/R$.

Lastly, to see that all homsets in $\mathbf{PER}$ and ω-$\mathbf{Sets}$ are countable, note that all the morphisms in these two categories are tracked by partial recursive functions, and there are only countably many partially recursive functions. $\square$

1.2.7.

(i) Prove that for each ω-set (I, E) the slice category ω-$\mathbf{Sets}/(I, E)$ is Cartesian closed, *i.e.* that ω-$\mathbf{Sets}$ is a locally Cartesian closed category (LCCC).

(ii) Show that also $\mathbf{PER}$ is an LCCC.

1.2.8. Show that a map $\nabla X \to (\mathbb{N}/R, \in)$ in ω-$\mathbf{Sets}$ is constant (where X is a set and R is a PER).

Solution. Let X be a set, R a PER, and $f : \nabla X \to (\mathbb{N}/R, \in)$ a morphism in ω-$\mathbf{Sets}$. Choose a code $e \in \mathbb{N}$ that tracks f. Then for all $x \in X$ and $n \in \mathbb{N}$, $e \cdot n$ is defined and $e \cdot n \in f(x) \in \mathbb{N}/R$. Since R is a PER, this implies that $f(x) = f(y)$ for all $x, y \in X$, so f is constant. $\square$

1.3 Some general examples

1.3.1.

(i) Show that in the total category $\mathrm{s}(\mathbb{B})$ of a simple fibration $\begin{array}{c}\mathrm{s}(\mathbb{B})\\ \downarrow\\ \mathbb{B}\end{array}$ a morphism $(u, f) : (I, X) \to (J, Y)$ is Cartesian if and only if there is an isomorphism $h : I \times X \xrightarrow{\cong} I \times Y$ in $\mathbb{B}$ such that $\pi \circ h = \pi$ and $\pi' \circ h = f$.

(ii) Show that the assignment $(I, X) \mapsto I^*(X) = \begin{pmatrix} I \times X \\ \downarrow_\pi \\ I \end{pmatrix}$ extends to a full and faithful functor $\mathrm{s}(\mathbb{B}) \to \mathbb{B}^{\to}$. Prove that it maps Cartesian morphisms to pullback squares.

[This functor restricts to a full and faithful functor $\mathbb{B} /\!\!/ I \to \mathbb{B}/I$.]

Remark 1.3.1.1. In part (i), the assertion that "there is an isomorphism $h : I \times X \to I \times Y$ in $\mathbb{B}$ such that $\pi \circ h = \pi$ and $\pi' \circ h = f$" is the same as saying that "$\langle \pi, f \rangle : I \times X \to I \times Y$ is an isomorphism."

Lemma 1.3.1.2. *Let $\mathbb{B}$ be a category with objects A, B, C, a product*

$$A \xleftarrow{\ \pi\ } A \times B \xrightarrow{\ \pi'\ } B$$

and morphisms $f : C \to A$ and $g : C \to B$ such that $\langle f, g \rangle : C \to A \times B$ is an isomorphism. Then

$$f \circ \langle f, g \rangle^{-1} = \pi, \qquad\qquad g \circ \langle f, g \rangle^{-1} = \pi'.$$

Proof. For the first case, we have

$$f \circ \langle f, g \rangle^{-1} = \pi \circ \langle f, g \rangle \circ \langle f, g \rangle^{-1} = \pi. \qquad \square$$

Solution, (i). ($\Longrightarrow$) Suppose $(u, f) : (I, X) \to (J, Y)$ is Cartesian in $\mathrm{s}(\mathbb{B})$ over $u : I \to J$. Consider the unique morphism $(\mathrm{id}, g) : (I, Y) \to (I, X)$ in $\mathrm{s}(\mathbb{B})$

over $\mathrm{id} : I \to I$ such that the following diagram commutes.

$$\begin{tikzcd}
(I,Y) \arrow[dr, dashed, "(\mathrm{id},g)"] \arrow[drr, bend left, "(u,\pi')"] \arrow[dd] & & \\
& (I,X) \arrow[r, "(u,f)"] \arrow[d] & (J,Y) \arrow[d] \\
I \arrow[r, "\mathrm{id}"'] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular,

$$\pi' = f \circ \langle \pi, g \rangle \,.$$

We will show that $\langle \pi, f \rangle : I \times X \to I \times Y$ is an isomorphism in $\mathbb{B}$ by showing that $\langle \pi, g \rangle : I \times Y \to I \times X$ is its inverse. On the one hand, we have

$$\begin{aligned}
\langle \pi, f \rangle \circ \langle \pi, g \rangle &= \langle \pi \circ \langle \pi, g \rangle \,, f \circ \langle \pi, g \rangle \rangle \\
&= \langle \pi, \pi' \rangle \\
&= \mathrm{id} \,.
\end{aligned}$$

On the other hand, we have

$$f = \pi' \circ \langle \pi, f \rangle \,,$$

so the upper triangle in the following diagram commutes, which makes the whole diagram commute.

$$\begin{tikzcd}
(I,X) \arrow[dr, "(\mathrm{id},f)"] \arrow[drrr, bend left, "(u,f)"] \arrow[dd] & & & \\
& (I,Y) \arrow[drr, bend left, "(u,\pi')"] \arrow[dr, "(\mathrm{id},g)"] \arrow[dd] & & \\
I \arrow[r, "\mathrm{id}"'] & I \arrow[dr, "\mathrm{id}"'] & (I,X) \arrow[r, "(u,f)"] \arrow[d] & (J,Y) \arrow[d] \\
& & I \arrow[r, "u"] & J
\end{tikzcd}$$

Since (u, f) is Cartesian, the morphism $(\mathrm{id}, g) \circ (\mathrm{id}, f)$ is the unique morphism $(I, X) \to (I, X)$ in $\mathrm{s}(\mathbb{B})$ over $\mathrm{id} : I \to I$ such that

$$(u, f) \circ (\mathrm{id}, g) \circ (\mathrm{id}, f) = (u, f),$$

but $\mathrm{id} = (\mathrm{id}, \pi') : (I, X) \to (I, X)$ is another such morphism, so

$$(\mathrm{id}, \pi') = (\mathrm{id}, g) \circ (\mathrm{id}, f).$$

In particular,

$$\pi' = g \circ \langle \pi, f \rangle ,$$

so

$$\begin{aligned} \langle \pi, g \rangle \circ \langle \pi, f \rangle &= \langle \pi \circ \langle \pi, f \rangle , g \circ \langle \pi, f \rangle \rangle \\ &= \langle \pi, \pi' \rangle \\ &= \mathrm{id} . \end{aligned}$$

Therefore, $\langle \pi, f \rangle : I \times X \to I \times Y$ is an isomorphism in $\mathbb{B}$.

($\Longleftarrow$) Let $(u, f) : (I, X) \to (J, Y)$ be a morphism in $\mathrm{s}(\mathbb{B})$ over $u : I \to J$, and suppose that $\langle \pi, f \rangle : I \times X \to I \times Y$ is an isomorphism. Consider another morphism $(v, g) : (K, Z) \to (J, Y)$ in $\mathrm{s}(\mathbb{B})$ and a morphism $w : K \to I$ in $\mathbb{B}$ such that $u \circ w = v$. This situation is summarized in the following commutative diagram.

$$\begin{tikzcd}
(K, Z) \arrow[rrr, bend left, "(v,g)"] \arrow[d] & & & \\
K \arrow[drr, bend left, "w"'] & (I, X) \arrow[r, "(u,f)"] \arrow[d] \arrow[drr, bend left, "v"] & (J, Y) \arrow[d] & \\
& I \arrow[r, "u"] & J &
\end{tikzcd}$$

Consider the following composition in $\mathbb{B}$ (call it $h : K \times Z \to X$).

$$K \times Z \xrightarrow{\langle w \circ \pi, g \rangle} I \times Y \xrightarrow{\langle \pi, f \rangle^{-1}} I \times X \xrightarrow{\pi'} X$$

We claim that $(w, h) : (K, Z) \to (I, X)$ is the unique morphism in $\mathrm{s}(\mathbb{B})$ such that $(u, f) \circ (w, h) = (v, g)$, which will imply that (u, f) is Cartesian. That

is, the following diagram will be shown to commute.

$$\begin{tikzcd}
(K,Z) \arrow[dr, dashed, "{(w,h)}"] \arrow[drr, bend left, "{(v,g)}"] \arrow[dd] & & \\
& (I,X) \arrow[r, "{(u,f)}"] \arrow[dd] & (J,Y) \arrow[dd] \\
K \arrow[dr, "w"'] \arrow[drr, bend left, "v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

We have $v = u \circ w$ by assumption, so to see that $(u, f) \circ (w, h) = (v, g)$, it remains to show that $g = f \circ \langle w \circ \pi, h \rangle$. Note that

$$\begin{aligned}
\pi \circ \langle \pi, f \rangle^{-1} \circ \langle w \circ \pi, g \rangle &= \pi \circ \langle w \circ \pi, g \rangle && \text{by Lemma 1.3.1.2} \\
&= w \circ \pi,
\end{aligned}$$

so the universal property of products yields

$$\begin{aligned}
\langle \pi, f \rangle^{-1} \circ \langle w \circ \pi, g \rangle &= \left\langle w \circ \pi, \pi' \circ \langle \pi, f \rangle^{-1} \circ \langle w \circ \pi, g \rangle \right\rangle \\
&= \langle w \circ \pi, h \rangle .
\end{aligned} \tag{1.3.1.1}$$

Then we have

$$\begin{aligned}
g &= \pi' \circ \langle w \circ \pi, g \rangle \\
&= f \circ \langle \pi, f \rangle^{-1} \circ \langle w \circ \pi, g \rangle && \text{by Lemma 1.3.1.2} \\
&= f \circ \langle w \circ \pi, h \rangle . && \text{by (1.3.1.1)}
\end{aligned}$$

To show uniqueness, suppose that $(w, h') : (K, Z) \to (I, X)$ is another morphism in $\mathrm{s}(\mathbb{B})$ over $w : K \to I$ such that $(u, f) \circ (w, h') = (v, g)$ (and, in particular, $g = f \circ \langle w \circ \pi, h' \rangle$). Then we have

$$\begin{aligned}
h' &= \pi' \circ \langle w \circ \pi, h' \rangle \\
&= \pi' \circ \langle \pi, f \rangle^{-1} \circ \langle \pi, f \rangle \circ \langle w \circ \pi, h' \rangle \\
&= \pi' \circ \langle \pi, f \rangle^{-1} \circ \langle \pi \circ \langle w \circ \pi, h' \rangle , f \circ \langle w \circ \pi, h' \rangle \rangle \\
&= \pi' \circ \langle \pi, f \rangle^{-1} \circ \langle w \circ \pi, g \rangle \\
&= h.
\end{aligned}$$

Therefore, $(u, f) : (I, X) \to (J, Y)$ is Cartesian in $\mathrm{s}(\mathbb{B})$ over $u : I \to J$. $\square$

1.6 Fibrations of signatures

1.6.1. Write down a (single-typed) signature for groups and also a (many-typed) signature for vector spaces.

Solution, groups. A possible single-typed signature for groups is given by a basic type G and function symbols

$$\begin{aligned} \mathsf{op} &: \mathsf{G}, \mathsf{G} \longrightarrow \mathsf{G}, \\ \mathsf{id} &: () \longrightarrow \mathsf{G}, \\ \mathsf{inv} &: \mathsf{G} \longrightarrow \mathsf{G}, \end{aligned}$$

representing the group operation, unit element, and inversion, respectively. □

Solution, vector spaces. It is not clear how to represent multiplicative inversion in a field—which is a partial function—so we will cheat and model (left) modules over a ring instead. A possible multi-typed signature for modules is given by two basic types R (the ring) and M (the abelian group on which the ring acts) and function symbols taken from the following collections.

(a) Function symbols representing the intrinsic ring attributes

$$\begin{aligned} \mathsf{add} &: \mathsf{R}, \mathsf{R} \longrightarrow \mathsf{R}, \\ \mathsf{multiply} &: \mathsf{R}, \mathsf{R} \longrightarrow \mathsf{R}, \\ \mathsf{0} &: () \longrightarrow \mathsf{R}, \\ \mathsf{1} &: () \longrightarrow \mathsf{R}, \\ \mathsf{negate} &: \mathsf{R} \longrightarrow \mathsf{R} \end{aligned}$$

(respectively, addition, multiplication, zero, one, and negation).

(b) Function symbols representing the intrinsic abelian group attributes

$$\begin{aligned} \mathsf{add} &: \mathsf{M}, \mathsf{M} \longrightarrow \mathsf{M}, \\ \mathsf{0} &: () \longrightarrow \mathsf{M}, \\ \mathsf{negate} &: \mathsf{M} \longrightarrow \mathsf{M} \end{aligned}$$

(respectively, addition, zero, and negation).

(c) A single function symbol

$$\mathsf{multiply} : \mathsf{R}, \mathsf{M} \longrightarrow \mathsf{M}$$

representing the action of the ring on the abelian group. □

1.6.2. A many-typed signature is called **finite** if it has only finitely many types and function symbols. Define the subcategory of **FinSign** $\hookrightarrow$ **Sign** consisting of finite signatures by change-of-base.

Solution. Recall that **Sign** can be defined via the change-of-base situation in the bottom right square of the following commutative diagram in **Cat**.

$$\begin{tikzcd}
 & & \mathrm{FinFam}(\mathbf{Sets}) \arrow[d, hook] \\
 & \mathbf{Sign} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Fam}(\mathbf{Sets}) \arrow[d] \\
\mathbf{FinSets} \arrow[r, hook] & \mathbf{Sets} \arrow[r, "T \mapsto T^* \times T"'] & \mathbf{Sets}
\end{tikzcd}$$

This situation can be completed by pullbacks in **Cat** as follows (noting that pullbacks of monic morphisms are monic; cf., e.g., [2, Proposition 2.5.3]).

$$\begin{tikzcd}
\mathbf{FinSign} \arrow[r, hook] \arrow[d, hook] \arrow[dr, phantom, "\lrcorner", very near start] & _ \arrow[r] \arrow[d, hook] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{FinFam}(\mathbf{Sets}) \arrow[d, hook] \\
_ \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbf{Sign} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Fam}(\mathbf{Sets}) \arrow[d] \\
\mathbf{FinSets} \arrow[r, hook] & \mathbf{Sets} \arrow[r, "T \mapsto T^* \times T"'] & \mathbf{Sets}
\end{tikzcd}$$

This presentation recovers **FinSign** as a subcategory of **Sign**, and the outer square is a pullback by Exercise 1.1.5, which simplifies to the following

change-of-base situation.

$$\begin{tikzcd} \mathbf{FinSign} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{FinFam}(\mathbf{Sets}) \arrow[d] \\ \mathbf{FinSets} \arrow[r, "T \mapsto T^* \times T"'] & \mathbf{Sets} \end{tikzcd}$$

□

1.6.3. Many-typed signatures are sometimes defined (like in [343] or in [282, 2.2.1]) as objects of the category $\mathbf{Sign}'$ which arises in the following change-of-base situation.

$$\begin{tikzcd} \mathbf{Sign}' \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbf{Sets}^{\to} \arrow[d, "\mathrm{cod}"] \\ \mathbf{Sets} \arrow[r, "T \mapsto T^* \times T"'] & \mathbf{Sets} \end{tikzcd}$$

(i) Describe the category $\mathbf{Sign}'$ in elementary terms.

(ii) Show that the categories $\mathbf{Sign}$ and $\mathbf{Sign}'$ are equivalent.

(iii) One often prefers $\mathbf{Sign}$ to $\mathbf{Sign}'$ because signatures in $\mathbf{Sign}$ allow **overloading** of function symbols: for example the use of $+$ both for addition of integers and for addition of reals. Explain.

[Another advantage of $\mathbf{Sign}$ is that $\begin{array}{c}\mathbf{Sign}\\ \downarrow \\ \mathbf{Sets}\end{array}$ is a *split* fibration.]

Solution, (i). The category $\mathbf{Sign}'$ has

objects Pairs $\Sigma = \langle T, \mathcal{F} \rangle$, where T is an ordinary set of **basic types** and

$$\mathcal{F} : X \longrightarrow T^* \times T,$$

is a "display-indexed" family of function symbols with total space X. If $f \in X$ and $\mathcal{F}(f) = \langle \langle \sigma_0, \ldots, \sigma_{n-1} \rangle, \sigma_n \rangle \in T^* \times T$, then f is a function

symbol with n inputs of types $\sigma_0, \ldots, \sigma_{n-1}$ and an output of type σ_n; this is written

$$f : \sigma_0,\ \ldots,\ \sigma_{n-1} \longrightarrow \sigma_n.$$

The fibre $\mathcal{F}^{-1}\left(\left\langle\left\langle\sigma_0, \ldots, \sigma_{n-1}\right\rangle, \sigma_n\right\rangle\right)$ over the index $\left\langle\left\langle\sigma_0, \ldots, \sigma_{n-1}\right\rangle, \sigma_n\right\rangle$ is the set of all such function symbols.

morphisms Given signatures $\Sigma = \langle T, \mathcal{F}\rangle$ and $\Sigma' = \langle S, \mathcal{G}\rangle$ in $\mathbf{Sign}'$, where

$$\mathcal{F} : X \longrightarrow T^* \times T, \qquad\qquad \mathcal{G} : Y \longrightarrow S^* \times S,$$

a morphism from Σ to Σ' is a pair $\langle u, \varphi\rangle$ where $u : T \to S$ and $\varphi : X \to Y$ are functions such that the following diagram commutes

$$\begin{tikzcd} X \arrow[r, "\varphi"] \arrow[d, "\mathcal{F}"'] & Y \arrow[d, "\mathcal{G}"] \\ T^* \times T \arrow[r, "u^* \times u"] & S^* \times S \end{tikzcd}$$

That is, if $f \in X$ with $f : \sigma_0,\ \ldots,\ \sigma_{n-1} \to \sigma_n$ (in Σ), then

$$\varphi(f) : u(\sigma_0),\ \ldots,\ u(\sigma_{n-1}) \longrightarrow u(\sigma_n) \qquad\qquad (\text{in } \Sigma') \qquad \square$$

Solution, (ii). The argument is nearly identical to the discussion preceding [7, Proposition 1.2.2] or the proof of Exercise 1.2.3(ii). $\square$

Solution, (iii). In **Sign**, function symbols are pointwise-indexed, so every index (i.e., finite list of input types together with an output type) has its own "namespace" of function symbols from which to choose, and the same function symbol can occur in the namespaces of multiple indices. In contrast, the function symbols in $\mathbf{Sign}'$ are taken from a global set and partitioned by index, so no function symbol can be assigned to more than one index. $\square$

1.6.4. Describe the category $\mathbf{S\text{-}Model_{ST}}$ of models of single-typed signatures in detail.

Solution. Recall ([7, Definition 1.6.7]) that the category $\mathbf{S\text{-}Model_{ST}}$ arises from the following stacked change-of-base situations.

$$\begin{tikzcd}
\mathbf{S\text{-}Model_{ST}} \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathbf{S\text{-}Model} \arrow[d] & \left(\Sigma, (A_\sigma)_{\sigma\in|\Sigma|}, [\![_]\!]\right) \arrow[d, mapsto] \\
\mathbf{Sign_{ST}} \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathbf{Sign} \arrow[d] & \Sigma \arrow[d, mapsto] \\
1 \arrow[r, hook, "_\mapsto 1"] & \mathbf{Sets} & |\Sigma|
\end{tikzcd}$$

For the "single-typed" left side of this diagram, choose a basic type $*$[1].

A single-typed signature Σ (in $\mathbf{Sign_{ST}}$) is then just a pointwise-indexed family of sets $(F_n)_{n\in\mathbb{N}}$, where each F_n is the set of function symbols with n inputs and one output (all having type $*$). That is, for $f \in F_n$ we have

$$f : \underbrace{*, \ldots, *}_{n \text{ times}} \longrightarrow *.$$

A morphism from $\Sigma = (F_n)_{n\in\mathbb{N}}$ to $\Sigma' = (F'_n)_{n\in\mathbb{N}}$ in $\mathbf{Sign_{ST}}$ is an $\mathbb{N}$-indexed family of functions $(\varphi_n : F_n \to F'_n)_{n\in\mathbb{N}}$, so that a function symbol f with n inputs in Σ yields a function symbol $\varphi_n(f)$ with n inputs in Σ'.

Finally, the category $\mathbf{S\text{-}Model_{ST}}$ of (set-theoretic) models of single-typed signatures consists of the following data.

objects Triples of the form $(\Sigma, A, [\![_]\!])$, where $\Sigma = (F_n)_{n\in\mathbb{N}}$ is a single-typed signature, A is the **carrier set** for the basic type $*$, and for each $n \in \mathbb{N}$ and function symbol $f \in F_n$,

$$[\![f]\!] : A^n \longrightarrow A.$$

morphisms A morphism

$$(\varphi, H) : (\Sigma, A, [\![_]\!]) \longrightarrow \left(\Sigma', A', [\![_]\!]'\right)$$

[1] This is essentially the same as working in an "untyped" system; see [7, Section 2.5] or [6, Section 21.4].

in $\mathbf{S\text{-}Model_{ST}}$ consists of a morphism of (single-typed) signatures $\varphi : \Sigma \to \Sigma'$ and a function $H : A \to A'$ such that the following diagram commutes for each n-ary function symbol f in Σ.

$$\begin{tikzcd} A^n \arrow[r, "H^n"] \arrow[d, "[\![f]\!]"'] & (A')^n \arrow[d, "[\![\varphi(f)]\!]'"] \\ A \arrow[r, "H"] & A' \end{tikzcd}$$

That is, for all $a_0, \ldots, a_{n-1} \in A$ we have

$$H\left([\![f]\!]\,(a_0, \ldots, a_{n-1})\right) = [\![\varphi(f)]\!]'\,(H(a_0), \ldots, H(a_{n-1}))\,. \qquad \square$$

1.6.5. The category **Sign** captures signatures of functions. A **signature of predicates** consists of a set of types T together with predicate symbols $R : \sigma_1, \ldots, \sigma_n$ where each σ_i is a type (element of T). Define an appropriate category of such signatures of predicates by change-of-base. Define also a category with both function and predicate symbols by change-of-base. [Such a category will be introduced in Definition 4.1.1.]

Solution. Recall that the category **Sign** of multi-typed function signatures can be defined by the following change of base situation.

$$\begin{tikzcd} \mathbf{Sign} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathrm{Fam}(\mathbf{Sets}) \arrow[d] \\ \mathbf{Sets} \arrow[r, "T \mapsto T^* \times T"] & \mathbf{Sets} \end{tikzcd}$$

In this case, the key for modeling function symbols is the functor $T \mapsto T^* \times T$ which represents a template for constructing function signatures with a finite list of inputs (T^*) and one output (T) out of a set T of basic types.

The analogous functors generating predicate signatures and function-and-predicate signatures, repsectively, are

$$T \longmapsto T^* \qquad\qquad T \longmapsto T^* \times (T+1) \cong (T^* \times T) + T^*.$$

These yield change-of-base situations that define categories $\mathbf{Sign}'$ and $\mathbf{Sign}''$ of multi-typed signatures of predicates and multi-typed signatures of functions and predicates, as follows.

$$\begin{tikzcd} \mathbf{Sign}' \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Fam}(\mathbf{Sets}) \arrow[d] \\ \mathbf{Sets} \arrow[r, "T \mapsto T^*"] & \mathbf{Sets} \end{tikzcd} \qquad \begin{tikzcd} \mathbf{Sign}'' \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Fam}(\mathbf{Sets}) \arrow[d] \\ \mathbf{Sets} \arrow[r, "T \mapsto T^* \times (T+1)"] & \mathbf{Sets} \end{tikzcd} \qquad \square$$

1.6.6. Let Σ be a signature and $T = |\Sigma|$ its set of types. We write $\mathbf{S\text{-}Model}(\Sigma)$ for the fibre category over Σ of the fibration $\begin{array}{c}\mathbf{S\text{-}Model}\\ \downarrow \\ \mathbf{Sign}\end{array}$.

This is the category of Σ-models.

(i) Show that the assignment

$$X = (X_\sigma)_{\sigma \in T} \longmapsto \Big(\mathrm{Terms}_\tau(X)\Big)_{\tau \in T}$$

extends to a functor from the category $\mathrm{Fam}\,(\mathbf{Sets})_T = \mathbf{Sets}^T$ of T-indexed families of sets to $\mathbf{S\text{-}Model}(\Sigma)$.

(ii) Assume $\Big((A_\sigma)_{\sigma \in T}\,, [\![_]\!]\Big)$ is a Σ-model. Verify that an interpretation $[\![_]\!]^\tau_\rho : \mathrm{Terms}_\tau(X) \to A_\tau$ is a morphism of Σ-models.

Solution, (i). We will define a "term model" functor

$$\mathrm{Terms} : \mathbf{Sets}^T \longrightarrow \mathbf{S\text{-}Model}(\Sigma)$$

First, consider the following "pre-functorial" data (assignments of objects and morphisms in $\mathbf{Sets}^T$ to objects and morphisms in $\mathbf{S\text{-}Model}(\Sigma)$).

objects For a T-indexed family of sets $X = (X_\sigma)_{\sigma \in T}$ (viewed as a set of variables for each of the basic types in T),

$$\begin{aligned} X \longmapsto & \Big((\mathrm{Terms}_\tau(X))_{\tau \in T}\,, [\![_]\!]_X\Big) \\ & = \text{the } \textbf{term model} \text{ of } \Sigma \text{ with variables from } X. \end{aligned}$$

That is, the right-hand side is the model of Σ whose **carrier sets** are $(\mathrm{Terms}_\tau(X))_{\tau\in T}$ and whose **interpretation function** $[\![_]\!]_X$ is given, for a function symbol $F : \tau_0, \ldots, \tau_{n-1} \to \tau_n$ in Σ, by

$$\begin{aligned}[\![F]\!]_X : \mathrm{Terms}_{\tau_0}(X) \times \cdots \times \mathrm{Terms}_{\tau_{n-1}}(X) &\longrightarrow \mathrm{Terms}_{\tau_n}(X)\\ (M_0, \ldots, M_{n-1}) &\longmapsto F(M_0, \ldots, M_{n-1}).\end{aligned}$$

morphisms For a morphism $f = (f_\sigma : X_\sigma \to Y_\sigma)_{\sigma\in T} : X \to Y$ in $\mathbf{Sets}^T$,

$$f \longmapsto [\![_]\!]_f : \Big((\mathrm{Terms}_\tau(X))_{\tau\in T}, [\![_]\!]_X\Big) \longrightarrow \Big((\mathrm{Terms}_\tau(Y))_{\tau\in T}, [\![_]\!]_Y\Big)$$

defined as follows. Viewing X as a set of variables used to interpret the term model $\Big((\mathrm{Terms}_\tau(Y))_{\tau\in T}, [\![_]\!]_Y\Big)$, define a **valuation**

$$\begin{aligned}\rho(f) &\coloneqq (\rho_\sigma(f) : X_\sigma \longrightarrow \mathrm{Terms}_\sigma(Y))_{\sigma\in T}\\ \rho_\sigma(f) &\coloneqq x \longmapsto f_\sigma(x) \qquad \text{(as a term in } \mathrm{Terms}_\sigma(Y)).\end{aligned}$$

This yields an interpretation

$$[\![_]\!]_f \coloneqq [\![_]\!]_{\rho(f)} = \Big([\![_]\!]^\tau_{\rho(f)} : \mathrm{Terms}_\tau(X) \longrightarrow \mathrm{Terms}_\tau(Y)\Big)_{\tau\in T},$$

which is a morphism of Σ-models from $\Big((\mathrm{Terms}_\tau(X))_{\tau\in T}, [\![_]\!]_X\Big)$ to $\Big((\mathrm{Terms}_\tau(Y))_{\tau\in T}, [\![_]\!]_Y\Big)$ by (ii).

It remains to verify that these assignments are functorial. First, induction on the structure of terms shows that

$$[\![t]\!]^\tau_{\rho(\mathrm{id}_X)} = t$$

for every term $t \in \mathrm{Terms}_\tau(X)$ and every basic type τ, so that

$$[\![_]\!]_{\mathrm{id}_X} = \mathrm{id}_{\left((\mathrm{Terms}_\tau(X))_{\tau\in T}, [\![_]\!]_X\right)}.$$

Second, for morphisms $f : X \to Y$ and $g : Y \to Z$ in $\mathbf{Sets}^T$, we have

$$[\![x]\!]^\tau_{\rho(g\circ f)} = (g \circ f)_\tau(x) = g_\tau(f_\tau(x)) = \Big[\!\Big[[\![x]\!]^\tau_{\rho(f)}\Big]\!\Big]^\tau_{\rho(g)} \qquad \text{for } x \in X_\tau$$

and, by induction on terms, for a function symbol $F : \tau_0, \ldots, \tau_{n-1} \to \tau_n$ in Σ and terms $M_0 \in \mathrm{Terms}_{\tau_0}(X), \ldots, M_{n-1} \in \mathrm{Terms}_{\tau_{n-1}}(X)$,

$$\begin{aligned}
[\![F(M_0, \ldots, M_{n-1})]\!]^{\tau_n}_{\rho(g\circ f)} &= [\![F]\!]_Z \left([\![M_0]\!]^{\tau_0}_{\rho(g\circ f)}, \ldots, [\![M_{n-1}]\!]^{\tau_{n-1}}_{\rho(g\circ f)} \right) \\
&= [\![F]\!]_Z \left(\left[\!\!\left[[\![M_0]\!]^{\tau_0}_{\rho(f)} \right]\!\!\right]^{\tau_0}_{\rho(g)}, \ldots, \left[\!\!\left[[\![M_{n-1}]\!]^{\tau_{n-1}}_{\rho(f)} \right]\!\!\right]^{\tau_{n-1}}_{\rho(g)} \right) \\
&= \left[\!\!\left[F([\![M_0]\!]^{\tau_0}_{\rho(f)}, \ldots, [\![M_{n-1}]\!]^{\tau_{n-1}}_{\rho(f)}) \right]\!\!\right]^{\tau_n}_{\rho(g)} \\
&= \left[\!\!\left[[\![F]\!]_Y \left([\![M_0]\!]^{\tau_0}_{\rho(f)}, \ldots, [\![M_{n-1}]\!]^{\tau_{n-1}}_{\rho(f)} \right) \right]\!\!\right]^{\tau_n}_{\rho(g)} \\
&= \left[\!\!\left[[\![F(M_0, \ldots, M_{n-1})]\!]^{\tau_n}_{\rho(f)} \right]\!\!\right]^{\tau_n}_{\rho(g)}.
\end{aligned}$$

We conclude that

$$[\![_]\!]_{g\circ f} = [\![_]\!]_g \circ [\![_]\!]_f$$

so $\mathrm{Terms} : \mathbf{Sets}^T \to \mathbf{S\text{-}Model}(\Sigma)$ is a functor. □

Solution, (ii). Let $\left((A_\sigma)_{\sigma\in T}, [\![_]\!] \right)$ be a Σ-model, $X = (X_\sigma)_{\sigma\in T}$ a T-indexed family of sets (viewed as sets of variables for the basic types in T), and $\rho = (\rho_\sigma : X_\sigma \to A_\sigma)_{\sigma\in T}$ a valuation. Then ρ induces an **interpretation**

$$[\![_]\!]_\rho \coloneqq \left([\![_]\!]^\tau_\rho : \mathrm{Terms}_\tau(X) \longrightarrow A_\tau \right)_{\tau\in T},$$

which is defined by induction on the structure of terms by

$$[\![x]\!]^\tau_\rho \coloneqq \rho_\tau(x)$$

for $x \in X_\tau$ and

$$[\![F(M_0, \ldots, M_{n-1})]\!]^{\tau_n}_\rho \coloneqq [\![F]\!] \left([\![M_0]\!]^{\tau_0}_\rho, \ldots, [\![M_{n-1}]\!]^{\tau_{n-1}}_\rho \right) \tag{1.6.6.1}$$

for function symbols $F : \tau_0, \ldots, \tau_{n-1} \to \tau_n$ in Σ and terms $M_i \in \mathrm{Terms}_{\tau_i}(X)$ ($i \in \{0, \ldots, n-1\}$). In fact, (1.6.6.1) immediately yields

$$\begin{aligned}
[\![F]\!] \left([\![M_0]\!]^{\tau_0}_\rho, \ldots, [\![M_{n-1}]\!]^{\tau_{n-1}}_\rho \right) &= [\![F(M_0, \ldots, M_{n-1})]\!]^{\tau_n}_\rho \\
&= [\![[\![F]\!]_X (M_0, \ldots, M_{n-1})]\!]^{\tau_n}_\rho.
\end{aligned}$$

In other words, the following diagram commutes for every function symbol $F : \tau_0, \ldots, \tau_{n-1} \to \tau_n$ in Σ, whence $[\![_]\!]_\rho$ is a morphism of Σ-models from the term model $\left((\mathrm{Terms}_\tau(X))_{\tau\in T}, [\![_]\!]_X\right)$ to $\left((A_\sigma)_{\sigma\in T}, [\![_]\!]\right)$.

$$\begin{tikzcd}
\mathrm{Terms}_{\tau_0}(X) \times \cdots \times \mathrm{Terms}_{\tau_{n-1}}(X) \arrow[r, "{[\![_]\!]_\rho^{\tau_0} \times \cdots \times [\![_]\!]_\rho^{\tau_{n-1}}}"] \arrow[d, "{[\![F]\!]_X}"'] & A_{\tau_0} \times \cdots \times A_{\tau_{n-1}} \arrow[d, "{[\![F]\!]}"] \\
\mathrm{Terms}_{\tau_n}(X) \arrow[r, "{[\![_]\!]_\rho^{\tau_n}}"] & A_{\tau_n}
\end{tikzcd}$$

□

1.6.7. Let $((A_\sigma), [\![_]\!])$ be a Σ-model and ρ a valuation $(X_\sigma \to A_\sigma)$.

(i) Show that
$$[\![M\,[N/x]]\!]_\rho = [\![M]\!]_{\rho(x\mapsto [\![N]\!]_\rho)}\,.$$

(ii) Let $((B_\sigma), [\![_]\!])$ be another Σ-model and $H_\sigma : A_\sigma \to B_\sigma$ be a morphism of Σ-models (*i.e.* a morphism in the fibre of **S-Model** over Σ). Show that
$$H\left([\![M]\!]_\rho\right) = [\![M]\!]_{H\circ\rho}\,.$$

Solution, (i). We proceed by induction on terms. First, for $\tau \in T$, $x, y \in X_\tau$, and $N \in \mathrm{Terms}_\tau(X)$,

$$\begin{aligned}
[\![y\,[N/x]]\!]_\rho^\tau &= \begin{cases} [\![N]\!]_\rho^\tau & \text{if } y = x \\ [\![y]\!]_\rho^\tau & \text{otherwise} \end{cases} \\
&= \begin{cases} [\![N]\!]_\rho^\tau & \text{if } y = x \\ \rho_\tau(y) & \text{otherwise} \end{cases} \\
&= \rho_\tau(x \longmapsto [\![N]\!]_\rho^\tau)(y) \\
&= [\![y]\!]_{\rho(x\mapsto [\![N]\!]_\rho^\tau)}^\tau\,.
\end{aligned}$$

Next, if $\tau_0, \ldots, \tau_n \in T$, $M_0 \in \mathrm{Terms}_{\tau_0}(X), \ldots, M_{n-1} \in \mathrm{Terms}_{\tau_{n-1}}(X)$, and

$F : \tau_0, \ldots, \tau_{n-1} \to \tau_n$ is a function symbol in Σ, then, by induction,

$$\begin{aligned}
&[\![F(M_0, \ldots, M_{n-1})\,[N/x]]\!]_\rho^{\tau_n} \\
&\quad = [\![F(M_0\,[N/x]\,, \ldots, M_{n-1}\,[N/x])]\!]_\rho^{\tau_n} \\
&\quad = [\![F]\!]\left([\![M_0\,[N/x]]\!]_\rho^{\tau_0}\,, \ldots, [\![M_{n-1}\,[N/x]]\!]_\rho^{\tau_{n-1}}\right) \\
&\quad = [\![F]\!]\left([\![M_0]\!]_{\rho(x\mapsto[\![N]\!]_\rho^\tau)}^{\tau_0}\,, \ldots, [\![M_{n-1}]\!]_{\rho(x\mapsto[\![N]\!]_\rho^\tau)}^{\tau_{n-1}}\right) \\
&\quad = [\![F(M_0, \ldots, M_{n-1})]\!]_{\rho(x\mapsto[\![N]\!]_\rho^\tau)}^{\tau_n}\,.
\end{aligned}$$

□

Solution, (ii). Let $\left((A_\sigma)_{\sigma\in T}\,, [\![_]\!]_A\right)$ and $\left((B_\sigma)_{\sigma\in T}\,, [\![_]\!]_B\right)$ be Σ-models, and let $H = (H_\sigma : A_\sigma \to B_\sigma)_{\sigma\in T}$ be a morphism of Σ-models. If $\tau \in T$ and $x \in X_\tau$, then

$$H_\tau\left([\![x]\!]_{A,\,\rho}^\tau\right) = H_\tau\left(\rho_\tau(x)\right) = [\![x]\!]_{B,\,H\circ\rho}^\tau\,.$$

Next, if $\tau_0, \ldots, \tau_n \in T$, $M_0 \in \mathrm{Terms}_{\tau_0}(X), \ldots, M_{n-1} \in \mathrm{Terms}_{\tau_{n-1}}(X)$, and $F : \tau_0, \ldots, \tau_{n-1} \to \tau_n$ is a function symbol in Σ, then, by induction,

$$\begin{aligned}
&H_{\tau_n}\left([\![F\,(M_0, \ldots, M_{n-1})]\!]_{A,\,\rho}^{\tau_n}\right) \\
&\quad = H_{\tau_n}\left([\![F]\!]_A\left([\![M_0]\!]_{A,\,\rho}^{\tau_0}\,, \ldots, [\![M_{n-1}]\!]_{A,\,\rho}^{\tau_{n-1}}\right)\right) \\
&\quad = [\![F]\!]_B\left(H_{\tau_0}\left([\![M_0]\!]_{A,\,\rho}^{\tau_0}\right), \ldots, H_{\tau_{n-1}}\left([\![M_{n-1}]\!]_{A,\,\rho}^{\tau_{n-1}}\right)\right) \\
&\quad = [\![F]\!]_B\left([\![M_0]\!]_{B,\,H\circ\rho}^{\tau_0}\,, \ldots, [\![M_{n-1}]\!]_{B,\,H\circ\rho}^{\tau_{n-1}}\right) \\
&\quad = [\![F\,(M_0, \ldots, M_{n-1})]\!]_{B,\,H\circ\rho}^{\tau_n}\,.
\end{aligned}$$

□

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